

Chapter 11

FUNDAMENTALS OF THERMAL RADIATION

Electromagnetic and Thermal Radiation

11-1C Electromagnetic waves are caused by accelerated charges or changing electric currents giving rise to electric and magnetic fields. Sound waves are caused by disturbances. Electromagnetic waves can travel in vacuum, sound waves cannot.

11-2C Electromagnetic waves are characterized by their frequency ν and wavelength λ . These two properties in a medium are related by $\lambda = c / \nu$ where c is the speed of light in that medium.

11-3C Visible light is a kind of electromagnetic wave whose wavelength is between 0.40 and 0.76 μm . It differs from the other forms of electromagnetic radiation in that it triggers the sensation of seeing in the human eye.

11-4C Infrared radiation lies between 0.76 and 100 μm whereas ultraviolet radiation lies between the wavelengths 0.01 and 0.40 μm . The human body does not emit any radiation in the ultraviolet region since bodies at room temperature emit radiation in the infrared region only.

11-5C Thermal radiation is the radiation emitted as a result of vibrational and rotational motions of molecules, atoms and electrons of a substance, and it extends from about 0.1 to 100 μm in wavelength. Unlike the other forms of electromagnetic radiation, thermal radiation is emitted by bodies because of their temperature.

11-6C Light (or visible) radiation consists of narrow bands of colors from violet to red. The color of a surface depends on its ability to reflect certain wavelength. For example, a surface that reflects radiation in the wavelength range 0.63-0.76 μm while absorbing the rest appears red to the eye. A surface that reflects all the light appears white while a surface that absorbs the entire light incident on it appears black. The color of a surface at room temperature is not related to the radiation it emits.

11-7C Radiation in opaque solids is considered surface phenomena since only radiation emitted by the molecules in a very thin layer of a body at the surface can escape the solid.

11-8C Because the snow reflects almost all of the visible and ultraviolet radiation, and the skin is exposed to radiation both from the sun and from the snow.

11-9C Microwaves in the range of 10^2 to 10^5 μm are very suitable for use in cooking as they are reflected by metals, transmitted by glass and plastics and absorbed by food (especially water) molecules. Thus the electric energy converted to radiation in a microwave oven eventually becomes part of the internal energy of the food with no conduction and convection thermal resistances involved. In conventional cooking, on the other hand, conduction and convection thermal resistances slow down the heat transfer, and thus the heating process.

11-10 Electricity is generated and transmitted in power lines at a frequency of 60 Hz. The wavelength of the electromagnetic waves is to be determined.

Analysis The wavelength of the electromagnetic waves is

$$\lambda = \frac{c}{\nu} = \frac{2.998 \times 10^8 \text{ m/s}}{60 \text{ Hz}(1/\text{s})} = \mathbf{4.997 \times 10^6 \text{ m}}$$

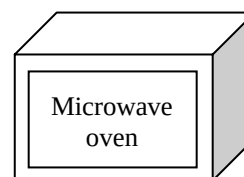
Power lines



11-11 A microwave oven operates at a frequency of 2.8×10^9 Hz. The wavelength of these microwaves and the energy of each microwave are to be determined.

Analysis The wavelength of these microwaves is

$$\lambda = \frac{c}{\nu} = \frac{2.998 \times 10^8 \text{ m/s}}{2.8 \times 10^9 \text{ Hz}(1/\text{s})} = 0.107 \text{ m} = \mathbf{107 \text{ mm}}$$



Then the energy of each microwave becomes

$$e = h\nu = \frac{hc}{\lambda} = \frac{(6.625 \times 10^{-34} \text{ Js})(2.998 \times 10^8 \text{ m/s})}{0.107 \text{ m}} = \mathbf{1.86 \times 10^{-24} \text{ J}}$$

11-12 A radio station is broadcasting radiowaves at a wavelength of 200 m. The frequency of these waves is to be determined.

Analysis The frequency of the waves is determined from

$$\lambda = \frac{c}{\nu} \longrightarrow \nu = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{200 \text{ m}} = \mathbf{1.5 \times 10^6 \text{ Hz}}$$



11-13 A cordless telephone operates at a frequency of 8.5×10^8 Hz. The wavelength of these telephone waves is to be determined.

Analysis The wavelength of the telephone waves is

$$\lambda = \frac{c}{\nu} = \frac{2.998 \times 10^8 \text{ m/s}}{8.5 \times 10^8 \text{ Hz}(1/\text{s})} = 0.35 \text{ m} = \mathbf{350 \text{ mm}}$$



Blackbody Radiation

11-14C A blackbody is a perfect emitter and absorber of radiation. A blackbody does not actually exist. It is an idealized body that emits the maximum amount of radiation that can be emitted by a surface at a given temperature.

11-15C *Spectral blackbody emissive power* is the amount of radiation energy emitted by a blackbody at an absolute temperature T per unit time, per unit surface area and per unit wavelength about wavelength λ . The integration of the spectral blackbody emissive power over the entire wavelength spectrum gives the *total blackbody emissive power*,

$$E_b(T) = \int_0^{\infty} E_{b\lambda}(T) d\lambda = \sigma T^4$$

The spectral blackbody emissive power varies with wavelength, the total blackbody emissive power does not.

11-16C We defined the blackbody radiation function f_λ because the integration $\int_0^{\infty} E_{b\lambda}(T) d\lambda$ cannot be performed. The blackbody radiation function f_λ represents the fraction of radiation emitted from a blackbody at temperature T in the wavelength range from $\lambda = 0$ to λ . This function is used to determine the fraction of radiation in a wavelength range between λ_1 and λ_2 .

11-17C The larger the temperature of a body, the larger the fraction of the radiation emitted in shorter wavelengths. Therefore, the body at 1500 K will emit more radiation in the shorter wavelength region. The body at 1000 K emits more radiation at $20 \mu m$ than the body at 1500 K since $\lambda T = \text{constant}$.

11-18 An isothermal cubical body is suspended in the air. The rate at which the cube emits radiation energy and the spectral blackbody emissive power are to be determined.

Assumptions The body behaves as a black body.

Analysis (a) The total blackbody emissive power is determined from Stefan-Boltzman Law to be

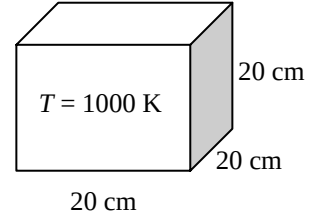
$$A_s = 6a^2 = 6(0.2^2) = 0.24 \text{ m}^2$$

$$E_b(T) = \sigma T^4 A_s = (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1000 \text{ K})^4 (0.24 \text{ m}^2) = \mathbf{1.36 \times 10^4 \text{ W}}$$

(b) The spectral blackbody emissive power at a wavelength of $4 \mu\text{m}$ is determined from Plank's distribution law,

$$E_{b\lambda} = \frac{C_1}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T}\right) - 1 \right]} = \frac{3.743 \times 10^8 \text{ W} \cdot \mu\text{m}^4 / \text{m}^2}{(4 \mu\text{m})^5 \left[\exp\left(\frac{1.4387 \times 10^4 \mu\text{m} \cdot \text{K}}{(4 \mu\text{m})(1000 \text{ K})}\right) - 1 \right]}$$

$$= \mathbf{10.3 \text{ kW/m}^2 \cdot \mu\text{m}}$$



11-19E The sun is at an effective surface temperature of 10,372 R. The rate of infrared radiation energy emitted by the sun is to be determined.

Assumptions The sun behaves as a black body.

Analysis Noting that $T = 10,400 \text{ R} = 5778 \text{ K}$, the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$ are determined from Table 11-2 to be

$$\lambda_1 T = (0.76 \mu\text{m})(5778 \text{ K}) = 4391.3 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.547370$$

$$\lambda_2 T = (100 \mu\text{m})(5778 \text{ K}) = 577,800 \mu\text{mK} \longrightarrow f_{\lambda_2} = 1.0$$

Then the fraction of radiation emitted between these two wavelengths becomes

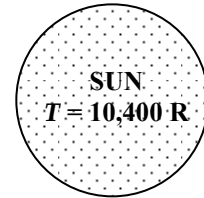
$$f_{\lambda_2} - f_{\lambda_1} = 1.0 - 0.547 = 0.453 \quad (\text{or } 45.3\%)$$

The total blackbody emissive power of the sun is determined from Stefan-Boltzman Law to be

$$E_b = \sigma T^4 = (0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4)(10,400 \text{ R})^4 = 2.005 \times 10^7 \text{ Btu/h} \cdot \text{ft}^2$$

Then,

$$E_{\text{infrared}} = (0.451)E_b = (0.453)(2.005 \times 10^7 \text{ Btu/h} \cdot \text{ft}^2) = \mathbf{9.08 \times 10^6 \text{ Btu/h} \cdot \text{ft}^2}$$



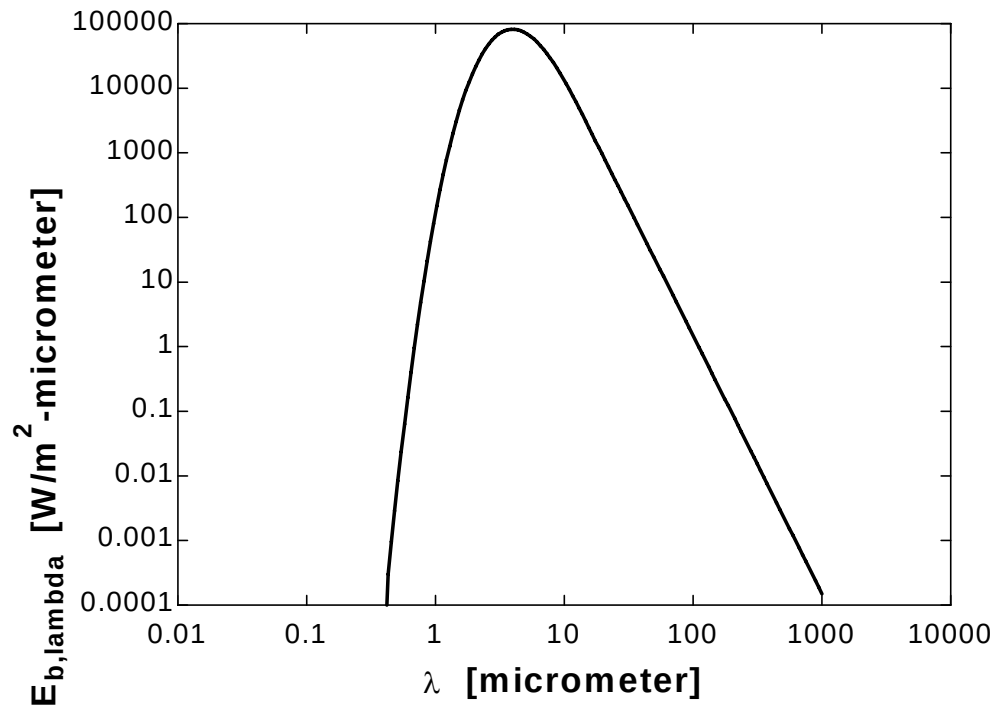
11-20E "PROBLEM 11-20"**"GIVEN"**

T=5780 "[K]"

"lambda=0.01[micrometer], parameter to be varied"

"ANALYSIS" $E_{b,\lambda} = C_1 / (\lambda^5 (\exp(C_2 / (\lambda T)) - 1))$ C₁=3.742E8 "[W-micrometer⁴/m²]"C₂=1.439E4 "[micrometer-K]"

λ [micrometer]	$E_{b,\lambda}$ [W/m ² -micrometer]
0.01	2.820E-90
10.11	12684
20.21	846.3
30.31	170.8
40.41	54.63
50.51	22.52
60.62	10.91
70.72	5.905
80.82	3.469
90.92	2.17
...	...
...	...
909.1	0.0002198
919.2	0.0002103
929.3	0.0002013
939.4	0.0001928
949.5	0.0001847
959.6	0.000177
969.7	0.0001698
979.8	0.0001629
989.9	0.0001563
1000	0.0001501



11-21 The temperature of the filament of an incandescent light bulb is given. The fraction of visible radiation emitted by the filament and the wavelength at which the emission peaks are to be determined.

Assumptions The filament behaves as a black body.

Analysis The visible range of the electromagnetic spectrum extends from $\lambda_1 = 0.40 \mu\text{m}$ to $\lambda_2 = 0.76 \mu\text{m}$. Noting that $T = 3200 \text{ K}$, the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$ are determined from Table 11-2 to be

$$\lambda_1 T = (0.40 \mu\text{m})(3200 \text{ K}) = 1280 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.0043964$$

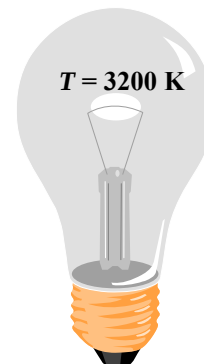
$$\lambda_2 T = (0.76 \mu\text{m})(3200 \text{ K}) = 2432 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.147114$$

Then the fraction of radiation emitted between these two wavelengths becomes

$$f_{\lambda_2} - f_{\lambda_1} = 0.14711424 - 0.0043964 = \mathbf{0.142718} \quad (\text{or } 14.3\%)$$

The wavelength at which the emission of radiation from the filament is maximum is

$$(\lambda T)_{\text{max power}} = 2897.8 \mu\text{m} \cdot \text{K} \longrightarrow \lambda_{\text{max power}} = \frac{2897.8 \mu\text{m} \cdot \text{K}}{3200 \text{ K}} = \mathbf{0.905 \mu\text{m}}$$



11-22 "PROBLEM 11-22"

"GIVEN"

"T=3200 [K], parameter to be varied"

lambda_1=0.40 "[micrometer]"

lambda_2=0.76 "[micrometer]"

"ANALYSIS"

$E_b \lambda = C_1 / (\lambda^5 (\exp(C_2 / (\lambda T)) - 1))$

$C_1 = 3.742 \times 10^8$ "[W-micrometer⁴/m²]"

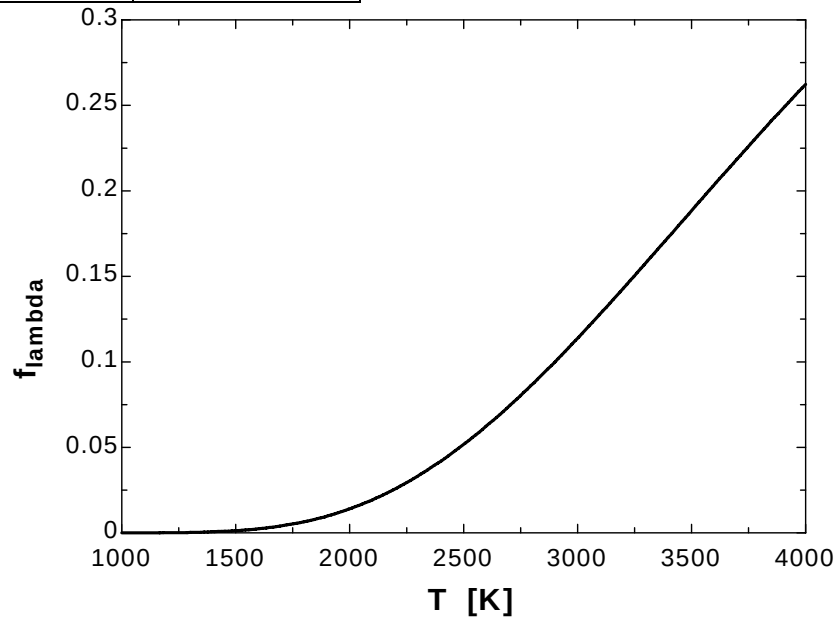
$C_2 = 1.439 \times 10^4$ "[micrometer-K]"

$f_\lambda = \text{integral}(E_b \lambda, \lambda, \lambda_1, \lambda_2) / E_b$

$E_b = \sigma T^4$

$\sigma = 5.67 \times 10^{-8}$ "[W/m²-K⁴], Stefan-Boltzmann constant"

T [K]	f_λ
1000	0.000007353
1200	0.0001032
1400	0.0006403
1600	0.002405
1800	0.006505
2000	0.01404
2200	0.02576
2400	0.04198
2600	0.06248
2800	0.08671
3000	0.1139
3200	0.143
3400	0.1732
3600	0.2036
3800	0.2336
4000	0.2623



11-23 An incandescent light bulb emits 15% of its energy at wavelengths shorter than $1\text{ }\mu\text{m}$. The temperature of the filament is to be determined.

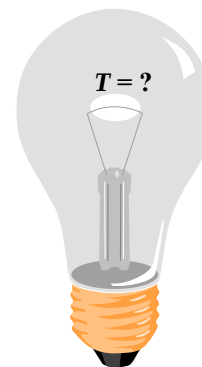
Assumptions The filament behaves as a black body.

Analysis From the Table 11-2 for the fraction of the radiation, we read

$$f_{\lambda} = 0.15 \longrightarrow \lambda T = 2445\text{ }\mu\text{mK}$$

For the wavelength range of $\lambda_1 = 0.0\text{ }\mu\text{m}$ to $\lambda_2 = 1.0\text{ }\mu\text{m}$

$$\lambda = 1\text{ }\mu\text{m} \longrightarrow \lambda T = 2445\text{ }\mu\text{mK} \longrightarrow T = \mathbf{2445\text{ K}}$$



11-24 Radiation emitted by a light source is maximum in the blue range. The temperature of this light source and the fraction of radiation it emits in the visible range are to be determined.

Assumptions The light source behaves as a black body.

Analysis The temperature of this light source is

$$(\lambda T)_{\max \text{ power}} = 2897.8 \mu\text{m} \cdot \text{K} \longrightarrow T = \frac{2897.8 \mu\text{m} \cdot \text{K}}{0.47 \mu\text{m}} = \mathbf{6166 \text{ K}}$$

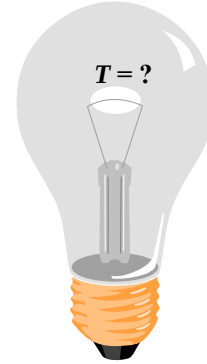
The visible range of the electromagnetic spectrum extends from $\lambda_1 = 0.40 \mu\text{m}$ to $\lambda_2 = 0.76 \mu\text{m}$. Noting that $T = 6166 \text{ K}$, the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$ are determined from Table 11-2 to be

$$\lambda_1 T = (0.40 \mu\text{m})(6166 \text{ K}) = 2466 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.15444$$

$$\lambda_2 T = (0.76 \mu\text{m})(6166 \text{ K}) = 4686 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.59141$$

Then the fraction of radiation emitted between these two wavelengths becomes

$$f_{\lambda_2} - f_{\lambda_1} = 0.59141 - 0.15444 \cong \mathbf{0.437} \quad (\text{or } 43.7\%)$$



11-25 A glass window transmits 90% of the radiation in a specified wavelength range and is opaque for radiation at other wavelengths. The rate of radiation transmitted through this window is to be determined for two cases.

Assumptions The sources behave as a black body.

Analysis The surface area of the glass window is

$$A_s = 4 \text{ m}^2$$

(a) For a blackbody source at 5800 K, the total blackbody radiation emission is

$$E_b(T) = \sigma T^4 A_s = (5.67 \times 10^{-8} \text{ kW/m}^2 \cdot \text{K}^4)(5800 \text{ K})^4 (4 \text{ m}^2) = 2.567 \times 10^5 \text{ kW}$$

The fraction of radiation in the range of 0.3 to 3.0 μm is

$$\lambda_1 T = (0.30 \mu\text{m})(5800 \text{ K}) = 1740 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.03345$$

$$\lambda_2 T = (3.0 \mu\text{m})(5800 \text{ K}) = 17,400 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.97875$$

$$\Delta f = f_{\lambda_2} - f_{\lambda_1} = 0.97875 - 0.03345 = 0.9453$$

Noting that 90% of the total radiation is transmitted through the window,

$$\begin{aligned} E_{\text{transmit}} &= 0.90 \Delta f E_b(T) \\ &= (0.90)(0.9453)(2.567 \times 10^5 \text{ kW}) = \mathbf{2.184 \times 10^5 \text{ kW}} \end{aligned}$$

(b) For a blackbody source at 1000 K, the total blackbody emissive power is

$$E_b(T) = \sigma T^4 A_s = (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1000 \text{ K})^4 (4 \text{ m}^2) = 226.8 \text{ kW}$$

The fraction of radiation in the visible range of 0.3 to 3.0 μm is

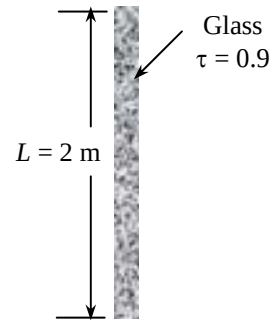
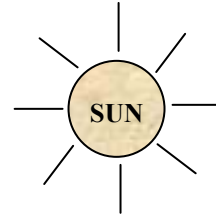
$$\lambda_1 T = (0.30 \mu\text{m})(1000 \text{ K}) = 300 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.0000$$

$$\lambda_2 T = (3.0 \mu\text{m})(1000 \text{ K}) = 3000 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.273232$$

$$\Delta f = f_{\lambda_2} - f_{\lambda_1} = 0.273232 - 0$$

and

$$E_{\text{transmit}} = 0.90 \Delta f E_b(T) = (0.90)(0.273232)(226.8 \text{ kW}) = \mathbf{55.8 \text{ kW}}$$



Radiation Intensity

11-26C A solid angle represents an opening in space, whereas a plain angle represents an opening in a plane. For a sphere of unit radius, the solid angle about the origin subtended by a given surface on the sphere is equal to the area of the surface. For a circle of unit radius, the plain angle about the origin subtended by a given arc is equal to the length of the arc. The value of a solid angle associated with a sphere is 4π .

11-27C The intensity of emitted radiation $I_e(\theta, \phi)$ is defined as the rate at which radiation energy $d\dot{Q}_e$ is emitted in the (θ, ϕ) direction per unit area normal to this direction and per unit solid angle about this direction. For a diffusely emitting surface, the emissive power is related to the intensity of emitted radiation by $E = \pi I_e$ (or $E_\lambda = \pi I_{\lambda,e}$ for spectral quantities).

11-28C Irradiation G is the radiation flux incident on a surface from all directions. For diffusely incident radiation, irradiation on a surface is related to the intensity of incident radiation by $G = \pi I_i$ (or $G_\lambda = \pi I_{\lambda,i}$ for spectral quantities).

11-29C Radiosity J is the rate at which radiation energy leaves a unit area of a surface by emission and reflection in all directions.. For a diffusely emitting and reflecting surface, radiosity is related to the intensity of emitted and reflected radiation by $J = \pi I_{e+r}$ (or $J_\lambda = \pi I_{\lambda,e+r}$ for spectral quantities).

11-30C When the variation of a spectral radiation quantity with wavelength is known, the corresponding total quantity is determined by integrating that quantity with respect to wavelength from $\lambda = 0$ to $\lambda = \infty$.

11-31 A surface is subjected to radiation emitted by another surface. The solid angle subtended and the rate at which emitted radiation is received are to be determined.

Assumptions 1 Surface A_1 emits diffusely as a blackbody. 2 Both A_1 and A_2 can be approximated as differential surfaces since both are very small compared to the square of the distance between them.

Analysis Approximating both A_1 and A_2 as differential surfaces, the solid angle subtended by A_2 when viewed from A_1 can be determined from Eq. 11-12 to be

$$\omega_{2-1} \cong \frac{A_{n,2}}{r^2} = \frac{A_2 \cos \theta_2}{r^2} = \frac{(4 \text{ cm}^2) \cos 60^\circ}{(80 \text{ cm})^2} = \mathbf{3.125 \times 10^{-4} \text{ sr}}$$

since the normal of A_2 makes 60° with the direction of viewing. Note that solid angle subtended by A_2 would be maximum if A_2 were positioned normal to the direction of viewing. Also, the point of viewing on A_1 is taken to be a point in the middle, but it can be any point since A_1 is assumed to be very small.

The radiation emitted by A_1 that strikes A_2 is equivalent to the radiation emitted by A_1 through the solid angle ω_{2-1} . The intensity of the radiation emitted by A_1 is

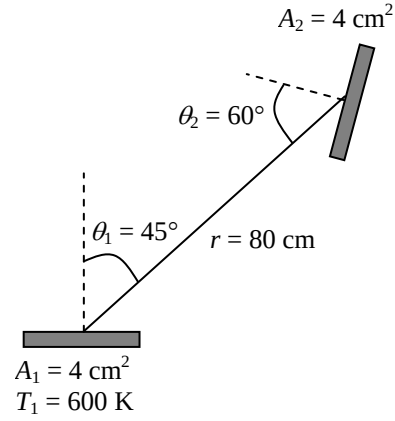
$$I_1 = \frac{E_b(T_1)}{\pi} = \frac{\sigma T_1^4}{\pi} = \frac{(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(800 \text{ K})^4}{\pi} = 7393 \text{ W/m}^2 \cdot \text{sr}$$

This value of intensity is the same in all directions since a blackbody is a diffuse emitter. Intensity represents the rate of radiation emission per unit area normal to the direction of emission per unit solid angle. Therefore, the rate of radiation energy emitted by A_1 in the direction of θ_1 through the solid angle ω_{2-1} is determined by multiplying I_1 by the area of A_1 normal to θ_1 and the solid angle ω_{2-1} . That is,

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (7393 \text{ W/m}^2 \cdot \text{sr})(4 \times 10^{-4} \text{ m}^2 \cos 45^\circ)(3.125 \times 10^{-4} \text{ sr}) \\ &= \mathbf{6.534 \times 10^{-4} \text{ W}} \end{aligned}$$

Therefore, the radiation emitted from surface A_1 will strike surface A_2 at a rate of $6.534 \times 10^{-4} \text{ W}$.

If A_2 were directly above A_1 at a distance 80 cm, $\theta_1 = 0^\circ$ and the rate of radiation energy emitted by A_1 becomes zero.



11-32 Radiation is emitted from a small circular surface located at the center of a sphere. Radiation energy streaming through a hole located on top of the sphere and the side of sphere are to be determined.

Assumptions 1 Surface A_1 emits diffusely as a blackbody. 2 Both A_1 and A_2 can be approximated as differential surfaces since both are very small compared to the square of the distance between them.

Analysis (a) Approximating both A_1 and A_2 as differential surfaces, the solid angle subtended by A_2 when viewed from A_1 can be determined from Eq. 11-12 to be

$$\omega_{2-1} \cong \frac{A_{n,2}}{r^2} = \frac{A_2}{r^2} = \frac{\pi(0.005 \text{ m})^2}{(1 \text{ m})^2} = 7.854 \times 10^{-5} \text{ sr}$$

since A_2 were positioned normal to the direction of viewing.

The radiation emitted by A_1 that strikes A_2 is equivalent to the radiation emitted by A_1 through the solid angle ω_{2-1} . The intensity of the radiation emitted by A_1 is

$$I_1 = \frac{E_b(T_1)}{\pi} = \frac{\sigma T_1^4}{\pi} = \frac{(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1000 \text{ K})^4}{\pi} = 18,048 \text{ W/m}^2 \cdot \text{sr}$$

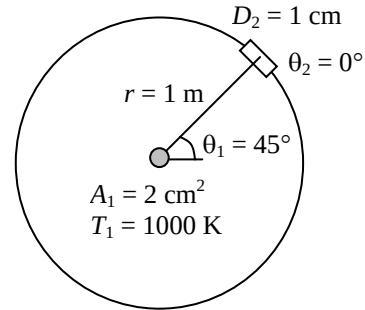
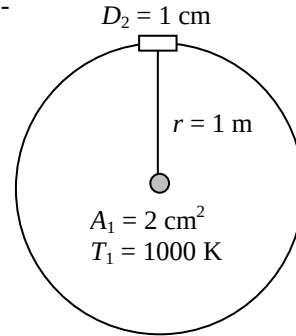
This value of intensity is the same in all directions since a blackbody is a diffuse emitter. Intensity represents the rate of radiation emission per unit area normal to the direction of emission per unit solid angle. Therefore, the rate of radiation energy emitted by A_1 in the direction of θ_1 through the solid angle ω_{2-1} is determined by multiplying I_1 by the area of A_1 normal to θ_1 and the solid angle ω_{2-1} . That is,

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (18,048 \text{ W/m}^2 \cdot \text{sr})(2 \times 10^{-4} \cos 0^\circ \text{ m}^2)(7.854 \times 10^{-5} \text{ sr}) \\ &= \mathbf{2.835 \times 10^{-4} \text{ W}} \end{aligned}$$

where $\theta_1 = 0^\circ$. Therefore, the radiation emitted from surface A_1 will strike surface A_2 at a rate of $2.835 \times 10^{-4} \text{ W}$.

(b) In this orientation, $\theta_1 = 45^\circ$ and $\theta_2 = 0^\circ$. Repeating the calculation we obtain the rate of radiation to be

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (18,048 \text{ W/m}^2 \cdot \text{sr})(2 \times 10^{-4} \cos 45^\circ \text{ m}^2)(7.854 \times 10^{-5} \text{ sr}) \\ &= \mathbf{2.005 \times 10^{-4} \text{ W}} \end{aligned}$$



11-33 Radiation is emitted from a small circular surface located at the center of a sphere. Radiation energy streaming through a hole located on top of the sphere and the side of sphere are to be determined.

Assumptions 1 Surface A_1 emits diffusely as a blackbody. 2 Both A_1 and A_2 can be approximated as differential surfaces since both are very small compared to the square of the distance between them.

Analysis (a) Approximating both A_1 and A_2 as differential surfaces, the solid angle subtended by A_2 when viewed from A_1 can be determined from Eq. 11-12 to be

$$\omega_{2-1} \cong \frac{A_{n,2}}{r^2} = \frac{A_2}{r^2} = \frac{\pi(0.005 \text{ m})^2}{(2 \text{ m})^2} = 1.963 \times 10^{-5} \text{ sr}$$

since A_2 were positioned normal to the direction of viewing.

The radiation emitted by A_1 that strikes A_2 is equivalent to the radiation emitted by A_1 through the solid angle ω_{2-1} . The intensity of the radiation emitted by A_1 is

$$I_1 = \frac{E_b(T_1)}{\pi} = \frac{\sigma T_1^4}{\pi} = \frac{(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1000 \text{ K})^4}{\pi} = 18,048 \text{ W/m}^2 \cdot \text{sr}$$

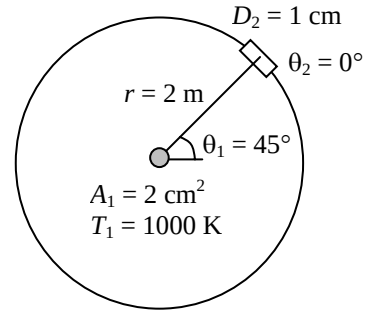
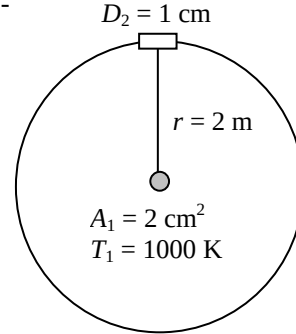
This value of intensity is the same in all directions since a blackbody is a diffuse emitter. Intensity represents the rate of radiation emission per unit area normal to the direction of emission per unit solid angle. Therefore, the rate of radiation energy emitted by A_1 in the direction of θ_1 through the solid angle ω_{2-1} is determined by multiplying I_1 by the area of A_1 normal to θ_1 and the solid angle ω_{2-1} . That is,

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (18,048 \text{ W/m}^2 \cdot \text{sr})(2 \times 10^{-4} \cos 0^\circ \text{ m}^2)(1.963 \times 10^{-5} \text{ sr}) \\ &= \mathbf{7.087 \times 10^{-5} \text{ W}} \end{aligned}$$

where $\theta_1 = 0^\circ$. Therefore, the radiation emitted from surface A_1 will strike surface A_2 at a rate of $2.835 \times 10^{-4} \text{ W}$.

(b) In this orientation, $\theta_1 = 45^\circ$ and $\theta_2 = 0^\circ$. Repeating the calculation we obtain the rate of radiation as

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (18,048 \text{ W/m}^2 \cdot \text{sr})(2 \times 10^{-4} \cos 45^\circ \text{ m}^2)(1.963 \times 10^{-5} \text{ sr}) \\ &= \mathbf{5.010 \times 10^{-5} \text{ W}} \end{aligned}$$



11-34 A small surface emits radiation. The rate of radiation energy emitted through a band is to be determined.

Assumptions Surface A emits diffusely as a blackbody.

Analysis The rate of radiation emission from a surface per unit surface area in the direction (θ, ϕ) is given as

$$dE = \frac{d\dot{Q}_e}{dA} = I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi$$

The total rate of radiation emission through the band between 60° and 45° can be expressed as

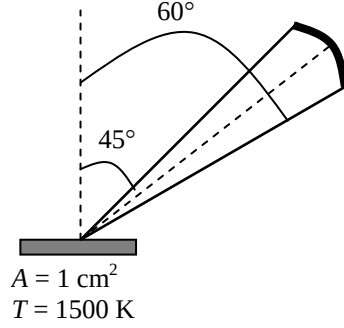
$$E = \int_{\phi=0}^{2\pi} \int_{\theta=45}^{60} I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi = I_b \frac{\pi}{4} = \frac{\sigma T^4}{\pi} \frac{\pi}{4} = \frac{\sigma T^4}{4}$$

since the blackbody radiation intensity is constant ($I_b = \text{constant}$), and

$$\int_{\phi=0}^{2\pi} \int_{\theta=45}^{60} \cos \theta \sin \theta d\theta d\phi = 2\pi \int_{\theta=45}^{60} \cos \theta \sin \theta d\theta = \pi(\sin^2 60 - \sin^2 45) = \pi/4$$

Approximating a small area as a differential area, the rate of radiation energy emitted from an area of 1 cm^2 in the specified band becomes

$$\dot{Q}_e = E dA = \frac{\sigma T^4}{4} dA = \frac{(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1500 \text{ K})^4}{4} (1 \times 10^{-4} \text{ m}^2) = \mathbf{7.18 \text{ W}}$$



11-35 A small surface is subjected to uniform incident radiation. The rates of radiation emission through two specified bands are to be determined.

Assumptions The intensity of incident radiation is constant.

Analysis (a) The rate at which radiation is incident on a surface per unit surface area in the direction (θ, ϕ) is given as

$$dG = \frac{d\dot{Q}_i}{dA} = I_i(\theta, \phi) \cos \theta \sin \theta d\theta d\phi$$

The total rate of radiation emission through the band between 0° and 45° can be expressed as

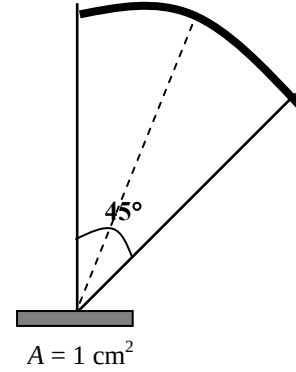
$$G_1 = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{45} I_i(\theta, \phi) \cos \theta \sin \theta d\theta d\phi = I_i \frac{\pi}{2}$$

since the incident radiation is constant ($I_i = \text{constant}$), and

$$\int_{\phi=0}^{2\pi} \int_{\theta=0}^{45} \cos \theta \sin \theta d\theta d\phi = 2\pi \int_{\theta=0}^{45} \cos \theta \sin \theta d\theta = \pi(\sin^2 45 - \sin^2 0) = \pi/2$$

Approximating a small area as a differential area, the rate of radiation energy emitted from an area of 1 cm^2 in the specified band becomes

$$\dot{Q}_{i,1} = G_1 dA = 0.5\pi I_i dA = 0.5\pi(2.2 \times 10^4 \text{ W/m}^2 \cdot \text{sr})(1 \times 10^{-4} \text{ m}^2) = \mathbf{3.46 \text{ W}}$$



(b) Similarly, the total rate of radiation emission through the band between 45° and 90° can be expressed as

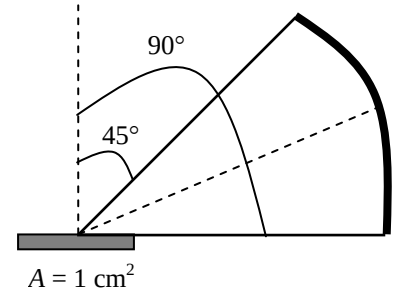
$$G_1 = \int_{\phi=0}^{2\pi} \int_{\theta=45}^{90} I_i(\theta, \phi) \cos \theta \sin \theta d\theta d\phi = I_i \frac{\pi}{2}$$

since

$$\int_{\phi=0}^{2\pi} \int_{\theta=45}^{90} \cos \theta \sin \theta d\theta d\phi = 2\pi \int_{\theta=45}^{90} \cos \theta \sin \theta d\theta = \pi(\sin^2 90 - \sin^2 45) = \pi/2$$

and

$$\dot{Q}_{i,2} = G_2 dA = 0.5\pi I_i dA = 0.5\pi(2.2 \times 10^4 \text{ W/m}^2 \cdot \text{sr})(1 \times 10^{-4} \text{ m}^2) = \mathbf{3.46 \text{ W}}$$



Discussion Note that the viewing area for the band $0^\circ - 45^\circ$ is much smaller, but the radiation energy incident through it is equal to the energy streaming through the remaining area.

Radiation Properties

11-36C The emissivity ε is the ratio of the radiation emitted by the surface to the radiation emitted by a blackbody at the same temperature. The fraction of radiation absorbed by the surface is called the absorptivity α ,

$$\varepsilon(T) = \frac{E(T)}{E_b(T)} \quad \text{and} \quad \alpha = \frac{\text{absorbed radiation}}{\text{incident radiation}} = \frac{G_{abs}}{G}$$

When the surface temperature is equal to the temperature of the source of radiation, the total hemispherical emissivity of a surface at temperature T is equal to its total hemispherical absorptivity for radiation coming from a blackbody at the same temperature $\varepsilon_\lambda(T) = \alpha_\lambda(T)$.

11-37C The fraction of irradiation reflected by the surface is called reflectivity ρ and the fraction transmitted is called the transmissivity τ

$$\rho = \frac{G_{ref}}{G} \quad \text{and} \quad \tau = \frac{G_{tr}}{G}$$

Surfaces are assumed to reflect in a perfectly spectral or diffuse manner for simplicity. In spectral (or mirror like) reflection, the angle of reflection equals the angle of incidence of the radiation beam. In diffuse reflection, radiation is reflected equally in all directions.

11-38C A body whose surface properties are independent of wavelength is said to be a graybody. The emissivity of a blackbody is one for all wavelengths, the emissivity of a graybody is between zero and one.

11-39C The heating effect which is due to the non-gray characteristic of glass, clear plastic, or atmospheric gases is known as the greenhouse effect since this effect is utilized primarily in greenhouses. The combustion gases such as CO_2 and water vapor in the atmosphere transmit the bulk of the solar radiation but absorb the infrared radiation emitted by the surface of the earth, acting like a heat trap. There is a concern that the energy trapped on earth will eventually cause global warming and thus drastic changes in weather patterns.

11-40C Glass has a transparent window in the wavelength range 0.3 to 3 μm and it is not transparent to the radiation which has wavelength range greater than 3 μm . Therefore, because the microwaves are in the range of 10^2 to 10^5 μm , the harmful microwave radiation cannot escape from the glass door.

11-41 The variation of emissivity of a surface at a specified temperature with wavelength is given. The average emissivity of the surface and its emissive power are to be determined.

Analysis The average emissivity of the surface can be determined from

$$\begin{aligned}\varepsilon(T) &= \frac{\varepsilon_1 \int_0^{\lambda_1} E_{b_\lambda}(T) d\lambda}{\sigma T^4} + \frac{\varepsilon_2 \int_{\lambda_1}^{\lambda_2} E_{b_\lambda}(T) d\lambda}{\sigma T^4} + \frac{\varepsilon_3 \int_{\lambda_2}^{\infty} E_{b_\lambda}(T) d\lambda}{\sigma T^4} \\ &= \varepsilon_1 f_{0-\lambda_1} + \varepsilon_2 f_{\lambda_1-\lambda_2} + \varepsilon_3 f_{\lambda_2-\infty} \\ &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (f_{\lambda_2} - f_{\lambda_1}) + \varepsilon_3 (1 - f_{\lambda_2})\end{aligned}$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$, determined from

$$\lambda_1 T = (2 \mu\text{m})(1000 \text{ K}) = 2000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.066728$$

$$\lambda_2 T = (6 \mu\text{m})(1000 \text{ K}) = 6000 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.737818$$

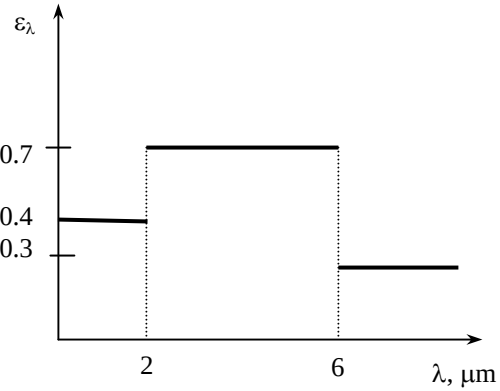
$$f_{0-\lambda_1} = f_{\lambda_1} - f_0 = f_{\lambda_1} \text{ since } f_0 = 0 \text{ and } f_{\lambda_2-\infty} = f_\infty - f_{\lambda_2} \text{ since } f_\infty = 1.$$

and,

$$\varepsilon = (0.4)(0.066728) + (0.7)(0.737818 - 0.066728) + (0.3)(1 - 0.737818) = \mathbf{0.575}$$

Then the emissive power of the surface becomes

$$E = \varepsilon \sigma T^4 = 0.575(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1000 \text{ K})^4 = \mathbf{32.6 \text{ kW/m}^2}$$

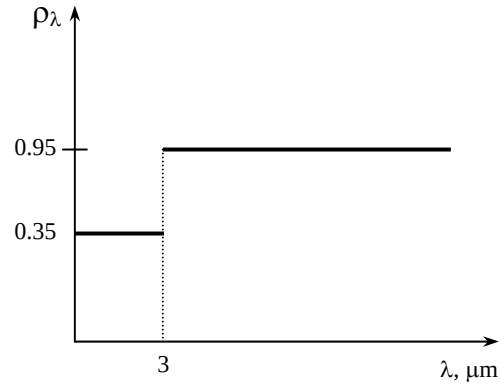


11-42 The variation of reflectivity of a surface with wavelength is given. The average reflectivity, emissivity, and absorptivity of the surface are to be determined for two source temperatures.

Analysis The average reflectivity of this surface for solar radiation ($T = 5800 \text{ K}$) is determined to be

$$\lambda T = (3 \mu\text{m})(5800 \text{ K}) = 17400 \mu\text{mK} \rightarrow f_{\lambda} = 0.978746$$

$$\begin{aligned} \rho(T) &= \rho_1 f_{0-\lambda_1}(T) + \rho_2 f_{\lambda_1-\infty}(T) \\ &= \rho_1 f_{\lambda_1} + \rho_2 (1 - f_{\lambda_1}) \\ &= (0.35)(0.978746) + (0.95)(1 - 0.978746) \\ &= \mathbf{0.362} \end{aligned}$$



Noting that this is an opaque surface, $\tau = 0$

$$\text{At } T = 5800 \text{ K: } \alpha + \rho = 1 \rightarrow \alpha = 1 - \rho = 1 - 0.362 = \mathbf{0.638}$$

Repeating calculations for radiation coming from surfaces at $T = 300 \text{ K}$,

$$\lambda T = (3 \mu\text{m})(300 \text{ K}) = 900 \mu\text{mK} \rightarrow f_{\lambda_1} = 0.0001685$$

$$\rho(T) = (0.35)(0.0001685) + (0.95)(1 - 0.0001685) = \mathbf{0.95}$$

$$\text{At } T = 300 \text{ K: } \alpha + \rho = 1 \rightarrow \alpha = 1 - \rho = 1 - 0.95 = \mathbf{0.05}$$

and $\varepsilon = \alpha = \mathbf{0.05}$

The temperature of the aluminum plate is close to room temperature, and thus emissivity of the plate will be equal to its absorptivity at room temperature. That is,

$$\varepsilon = \varepsilon_{\text{room}} = 0.05$$

$$\alpha = \alpha_s = 0.638$$

which makes it suitable as a solar collector. ($\alpha_s = 1$ and $\varepsilon_{\text{room}} = 0$ for an ideal solar collector)

11-43 The variation of transmissivity of the glass window of a furnace at a specified temperature with wavelength is given. The fraction and the rate of radiation coming from the furnace and transmitted through the window are to be determined.

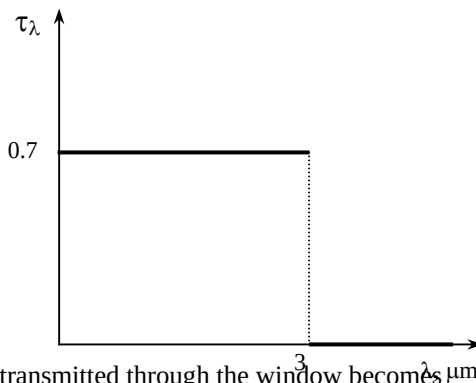
Assumptions The window glass behaves as a black body.

Analysis The fraction of radiation at wavelengths smaller than $3 \mu\text{m}$ is

$$\lambda T = (3 \mu\text{m})(1200 \text{ K}) = 3600 \mu\text{mK} \rightarrow f_{\lambda} = 0.403607$$

The fraction of radiation coming from the furnace and transmitted through the window is

$$\begin{aligned} \tau(T) &= \tau_1 f_{\lambda} + \tau_2 (1 - f_{\lambda}) \\ &= (0.7)(0.403607) + (0)(1 - 0.403607) \\ &= \mathbf{0.283} \end{aligned}$$



Then the rate of radiation coming from the furnace and transmitted through the window becomes μm

$$G_{tr} = \tau A \sigma T^4 = 0.283(0.25 \times 0.25 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1200 \text{ K})^4 = \mathbf{2076 \text{ W}}$$

11-44 The variation of emissivity of a tungsten filament with wavelength is given. The average emissivity, absorptivity, and reflectivity of the filament are to be determined for two temperatures.

Analysis (a) $T = 2000$ K

$$\lambda_1 T = (1 \mu\text{m})(2000 \text{ K}) = 2000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.066728$$

The average emissivity of this surface is

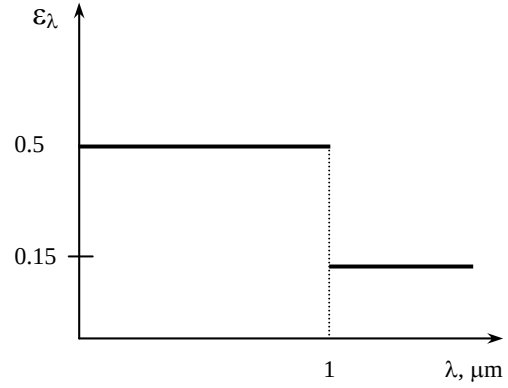
$$\begin{aligned} \varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.5)(0.066728) + (0.15)(1 - 0.066728) \\ &= \mathbf{0.173} \end{aligned}$$

From Kirchhoff's law,

$$\varepsilon = \alpha = \mathbf{0.173} \quad (\text{at } 2000 \text{ K})$$

and

$$\alpha + \rho = 1 \longrightarrow \rho = 1 - \alpha = 1 - 0.173 = \mathbf{0.827}$$



(b) $T = 3000$ K

$$\lambda_1 T = (1 \mu\text{m})(3000 \text{ K}) = 3000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.273232$$

Then

$$\begin{aligned} \varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.5)(0.273232) + (0.15)(1 - 0.273232) \\ &= \mathbf{0.246} \end{aligned}$$

From Kirchhoff's law,

$$\varepsilon = \alpha = \mathbf{0.246} \quad (\text{at } 3000 \text{ K})$$

and

$$\alpha + \rho = 1 \longrightarrow \rho = 1 - \alpha = 1 - 0.246 = \mathbf{0.754}$$

11-45 The variations of emissivity of two surfaces are given. The average emissivity, absorptivity, and reflectivity of each surface are to be determined at the given temperature.

Analysis For the first surface:

$$\lambda_1 T = (3 \mu\text{m})(3000 \text{ K}) = 9000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.890029$$

The average emissivity of this surface is

$$\begin{aligned}\varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.2)(0.890029) + (0.9)(1 - 0.890029) \\ &= \mathbf{0.28}\end{aligned}$$

The absorptivity and reflectivity are determined from Kirchhoff's law

$$\begin{aligned}\varepsilon &= \alpha = \mathbf{0.28} \quad (\text{at } 3000 \text{ K}) \\ \alpha + \rho &= 1 \longrightarrow \rho = 1 - \alpha = 1 - 0.28 = \mathbf{0.72}\end{aligned}$$

For the second surface:

$$\lambda_1 T = (3 \mu\text{m})(3000 \text{ K}) = 9000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.890029$$

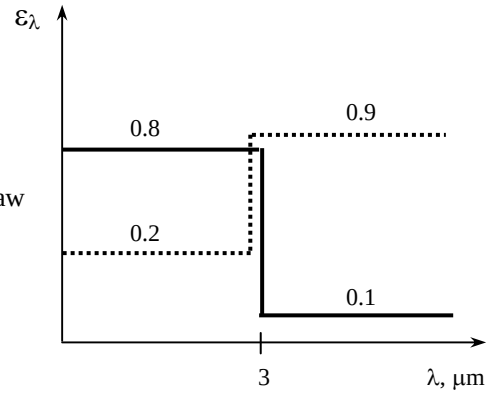
The average emissivity of this surface is

$$\begin{aligned}\varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.8)(0.890029) + (0.1)(1 - 0.890029) \\ &= \mathbf{0.72}\end{aligned}$$

Then,

$$\begin{aligned}\varepsilon &= \alpha = \mathbf{0.72} \quad (\text{at } 3000 \text{ K}) \\ \alpha + \rho &= 1 \longrightarrow \rho = 1 - \alpha = 1 - 0.72 = \mathbf{0.28}\end{aligned}$$

Discussion The second surface is more suitable to serve as a solar absorber since its absorptivity for short wavelength radiation (typical of radiation emitted by a high-temperature source such as the sun) is high, and its emissivity for long wavelength radiation (typical of emitted radiation from the absorber plate) is low.



11-46 The variation of emissivity of a surface with wavelength is given. The average emissivity and absorptivity of the surface are to be determined for two temperatures.

Analysis (a) For $T = 5800$ K:

$$\lambda_1 T = (5 \mu\text{m})(5800 \text{ K}) = 29,000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.99534$$

The average emissivity of this surface is

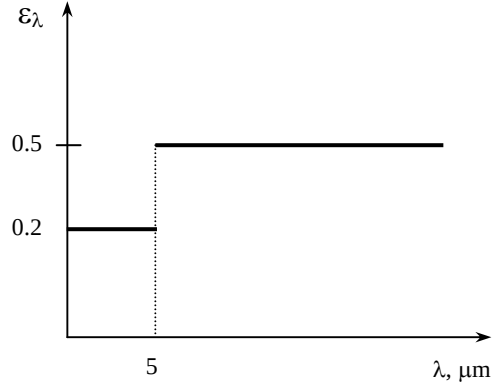
$$\begin{aligned}\varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.2)(0.99534) + (0.9)(1 - 0.99534) \\ &= \mathbf{0.203}\end{aligned}$$

(b) For $T = 300$ K:

$$\lambda_1 T = (5 \mu\text{m})(300 \text{ K}) = 1500 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.013754$$

and

$$\begin{aligned}\varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.2)(0.013754) + (0.9)(1 - 0.013754) \\ &= \mathbf{0.89}\end{aligned}$$



The absorptivities of this surface for radiation coming from sources at 5800 K and 300 K are, from Kirchhoff's law,

$$\alpha = \varepsilon = \mathbf{0.203} \quad (\text{at } 5800 \text{ K})$$

$$\alpha = \varepsilon = \mathbf{0.89} \quad (\text{at } 300 \text{ K})$$

11-47 The variation of absorptivity of a surface with wavelength is given. The average absorptivity, reflectivity, and emissivity of the surface are to be determined at given temperatures.

Analysis For $T = 2500$ K:

$$\lambda_1 T = (2 \mu\text{m})(2500 \text{ K}) = 5000 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.633747$$

The average absorptivity of this surface is

$$\begin{aligned}\alpha(T) &= \alpha_1 f_{\lambda_1} + \alpha_2 (1 - f_{\lambda_1}) \\ &= (0.2)(0.633747) + (0.7)(1 - 0.633747) \\ &= \mathbf{0.38}\end{aligned}$$

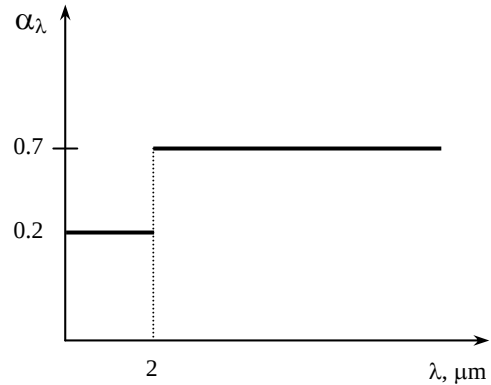
Then the reflectivity of this surface becomes

$$\alpha + \rho = 1 \longrightarrow \rho = 1 - \alpha = 1 - 0.38 = \mathbf{0.62}$$

Using Kirchhoff's law, $\alpha = \varepsilon$, the average emissivity of this surface at $T = 3000$ K is determined to be

$$\lambda T = (2 \mu\text{m})(3000 \text{ K}) = 6000 \mu\text{mK} \longrightarrow f_{\lambda} = 0.737818$$

$$\begin{aligned}\varepsilon(T) &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (1 - f_{\lambda_1}) \\ &= (0.2)(0.737818) + (0.7)(1 - 0.737818) \\ &= \mathbf{0.33}\end{aligned}$$



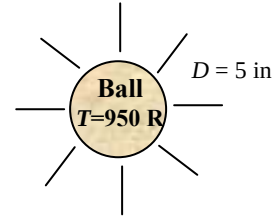
11-48E A spherical ball emits radiation at a certain rate. The average emissivity of the ball is to be determined at the given temperature.

Analysis The surface area of the ball is

$$A = \pi D^2 = \pi(5/12 \text{ ft})^2 = 0.5454 \text{ ft}^2$$

Then the average emissivity of the ball at this temperature is determined to be

$$E = \varepsilon A \sigma T^4 \longrightarrow \varepsilon = \frac{E}{A \sigma T^4} = \frac{120 \text{ Btu/h}}{(0.5454 \text{ ft}^2)(0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4)(950 \text{ R})^4} = \mathbf{0.158}$$



11-49 The variation of transmissivity of a glass is given. The average transmissivity of the pane at two temperatures and the amount of solar radiation transmitted through the pane are to be determined.

Analysis For $T=5800 \text{ K}$:

$$\lambda_1 T_1 = (0.3 \mu\text{m})(5800 \text{ K}) = 1740 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.035$$

$$\lambda_2 T_1 = (3 \mu\text{m})(5800 \text{ K}) = 17,400 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.977$$

The average transmissivity of this surface is

$$\tau(T) = \tau_1(f_{\lambda_2} - f_{\lambda_1}) = (0.9)(0.977 - 0.035) = \mathbf{0.848}$$

For $T=300 \text{ K}$:

$$\lambda_1 T_2 = (0.3 \mu\text{m})(300 \text{ K}) = 90 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.0$$

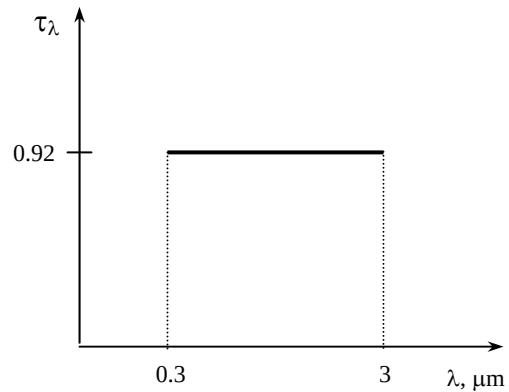
$$\lambda_2 T_2 = (3 \mu\text{m})(300 \text{ K}) = 900 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.0001685$$

Then,

$$\tau(T) = \tau_1(f_{\lambda_2} - f_{\lambda_1}) = (0.9)(0.0001685 - 0.0) = \mathbf{0.00015 \approx 0}$$

The amount of solar radiation transmitted through this glass is

$$G_{\text{tr}} = \tau G_{\text{incident}} = 0.848(650 \text{ W/m}^2) = \mathbf{551 \text{ W/m}^2}$$



Atmospheric and Solar Radiation

11-50C The solar constant represents the rate at which solar energy is incident on a surface normal to sun's rays at the outer edge of the atmosphere when the earth is at its mean distance from the sun. Its value is $G_s = 1353 \text{ W/m}^2$. The solar constant is used to estimate the effective surface temperature of the sun from the requirement that

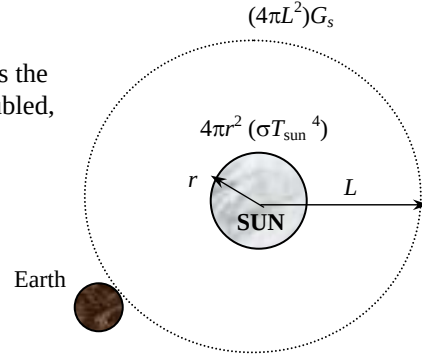
$$(4\pi L^2)G_{s1} = (4\pi r^2)\sigma T_{\text{sun}}^4$$

where L is the mean distance between the sun and the earth and r is the radius of the sun. If the distance between the earth and the sun doubled, the value of G_s drops to one-fourth since

$$4\pi(2L)^2 G_{s2} = (4\pi r^2)\sigma T_{\text{sun}}^4$$

$$16\pi L^2 G_{s2} = (4\pi r^2)\sigma T_{\text{sun}}^4$$

$$16\pi L^2 G_{s2} = 4\pi L^2 G_{s1} \longrightarrow G_{s2} = \frac{G_{s1}}{4}$$



11-51C The amount of solar radiation incident on earth will decrease by a factor of

$$\text{Reduction factor} = \frac{\sigma T_{\text{sun}}^4}{\sigma T_{\text{sun,new}}^4} = \frac{5762^4}{2000^4} = 68.9$$

(or to 1.5% of what it was). Also, the fraction of radiation in the visible range would be much smaller.

11-52C Air molecules scatter blue light much more than they do red light. This molecular scattering in all directions is what gives the sky its bluish color. At sunset, the light travels through a thicker layer of atmosphere, which removes much of the blue from the natural light, letting the red dominate.

11-53C The reason for different seasons is the tilt of the earth which causes the solar radiation to travel through a longer path in the atmosphere in winter, and a shorter path in summer. Therefore, the solar radiation is attenuated much more strongly in winter.

11-54C The gas molecules and the suspended particles in the atmosphere emit radiation as well as absorbing it. Although this emission is far from resembling the distribution of radiation from a blackbody, it is found convenient in radiation calculations to treat the atmosphere as a blackbody at some lower fictitious temperature that emits an equivalent amount of radiation energy. This fictitious temperature is called the effective sky temperature T_{sky} .

11-55C There is heat loss from both sides of the bridge (top and bottom surfaces of the bridge) which reduces temperature of the bridge surface to very low values. The relatively warm earth under a highway supply heat to the surface continuously, making the water on it less likely to freeze.

11-56C Due to its nearly horizontal orientation, windshield exchanges heat with the sky that is at very low temperature. Side windows on the other hand exchange heat with surrounding surfaces that are at relatively high temperature.

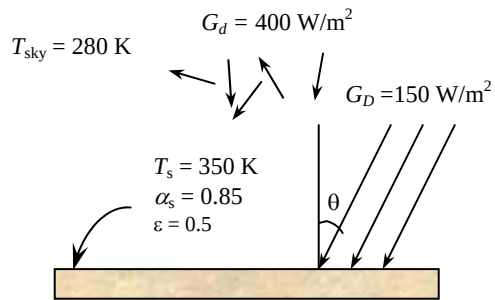
11-57C Because of different wavelengths of solar radiation and radiation originating from surrounding bodies, the surfaces usually have quite different absorptivities. Solar radiation is concentrated in the short wavelength region and the surfaces in the infrared region.

11-58 A surface is exposed to solar and sky radiation. The net rate of radiation heat transfer is to be determined.

Properties The solar absorptivity and emissivity of the surface are given to $\alpha_s = 0.85$ and $\varepsilon = 0.5$.

Analysis The total solar energy incident on the surface is

$$\begin{aligned} G_{\text{solar}} &= G_D \cos \theta + G_d \\ &= (350 \text{ W/m}^2) \cos 30^\circ + (400 \text{ W/m}^2) \\ &= 703.1 \text{ W/m}^2 \end{aligned}$$



Then the net rate of radiation heat transfer in this case becomes

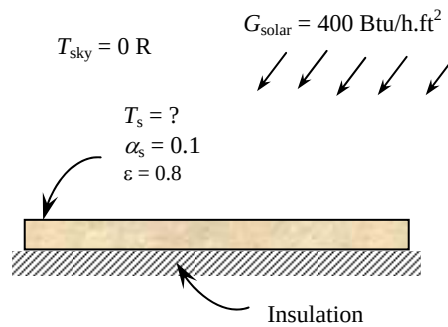
$$\begin{aligned} \dot{q}_{\text{net,rad}} &= \alpha_s G_{\text{solar}} - \varepsilon \sigma (T_s^4 - T_{\text{sky}}^4) \\ &= 0.85(703.1 \text{ W/m}^2) - 0.5(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(350 \text{ K})^4 - (280 \text{ K})^4] \\ &= \mathbf{347 \text{ W/m}^2} \text{ (to the surface)} \end{aligned}$$

11-59E A surface is exposed to solar and sky radiation. The equilibrium temperature of the surface is to be determined.

Properties The solar absorptivity and emissivity of the surface are given to $\alpha_s = 0.10$ and $\varepsilon = 0.8$.

Analysis The equilibrium temperature of the surface in this case is

$$\begin{aligned} \dot{q}_{\text{net,rad}} &= \alpha_s G_{\text{solar}} - \varepsilon \sigma (T_s^4 - T_{\text{sky}}^4) = 0 \\ \alpha_s G_{\text{solar}} &= \varepsilon \sigma (T_s^4 - T_{\text{sky}}^4) \\ 0.1(400 \text{ Btu/h.ft}^2) &= 0.8(0.1714 \times 10^{-8} \text{ Btu/h.ft}^2 \cdot \text{R}^4) [T_s^4 - (0 \text{ R})^4] \\ T_s &= \mathbf{413.3 \text{ R}} \end{aligned}$$



11-60 Water is observed to have frozen one night while the air temperature is above freezing temperature. The effective sky temperature is to be determined.

Properties The emissivity of water is $\varepsilon = 0.95$ (Table A-21).

Analysis Assuming the water temperature to be 0°C , the value of the effective sky temperature is determined from an energy balance on water to be

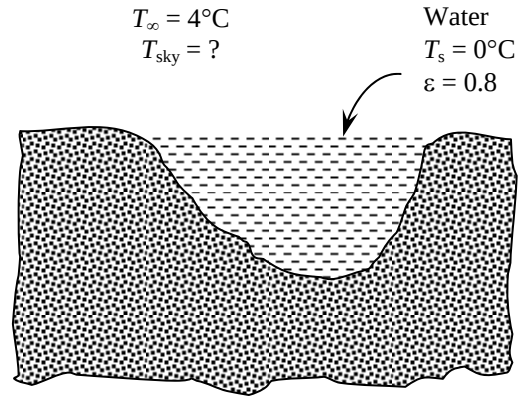
$$h(T_{\text{air}} - T_{\text{surface}}) = \varepsilon\sigma(T_s^4 - T_{\text{sky}}^4)$$

and

$$(18 \text{ W/m}^2 \cdot ^\circ\text{C})(4^\circ\text{C} - 0^\circ\text{C}) = 0.95(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(273 \text{ K})^4 - T_{\text{sky}}^4]$$

$$\longrightarrow T_{\text{sky}} = \mathbf{254.8 \text{ K}}$$

Therefore, the effective sky temperature must have been below 255 K.



11-61 The absorber plate of a solar collector is exposed to solar and sky radiation. The net rate of solar energy absorbed by the absorber plate is to be determined.

Properties The solar absorptivity and emissivity of the surface are given to $\alpha_s = 0.87$ and $\varepsilon = 0.09$.

Analysis The net rate of solar energy delivered by the absorber plate to the water circulating behind it can be determined from an energy balance to be

$$\dot{q}_{\text{net}} = \dot{q}_{\text{gain}} - \dot{q}_{\text{loss}}$$

$$\dot{q}_{\text{net}} = \alpha_s G_{\text{solar}} - [\varepsilon\sigma(T_s^4 - T_{\text{sky}}^4) + h(T_s - T_{\text{air}})]$$

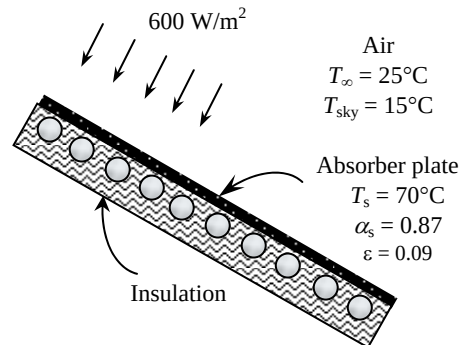
Then,

$$\dot{q}_{\text{net}} = 0.87(600 \text{ W/m}^2) - 0.09(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(70 + 273 \text{ K})^4 - (15 + 273 \text{ K})^4]$$

$$- (10 \text{ W/m}^2 \cdot \text{K})(70^\circ\text{C} - 25^\circ\text{C})$$

$$= \mathbf{36.5 \text{ W/m}^2}$$

Therefore, heat is gained by the plate and transferred to water at a rate of 36.5 W per m^2 surface area.



11-62 "PROBLEM 11-62"

"GIVEN"

" $\alpha_s=0.87$ parameter to be varied"

$\epsilon=0.09$

$G_{\text{solar}}=600 \text{ [W/m}^2\text{]}$

$T_{\text{air}}=25+273 \text{ [K]}$

$T_{\text{sky}}=15+273 \text{ [K]}$

$T_s=70+273 \text{ [K]}$

$h=10 \text{ [W/m}^2\text{-K]}$

$\sigma=5.67\text{E-}8 \text{ [W/m}^2\text{-K}^4\text{], Stefan-Boltzmann constant}$

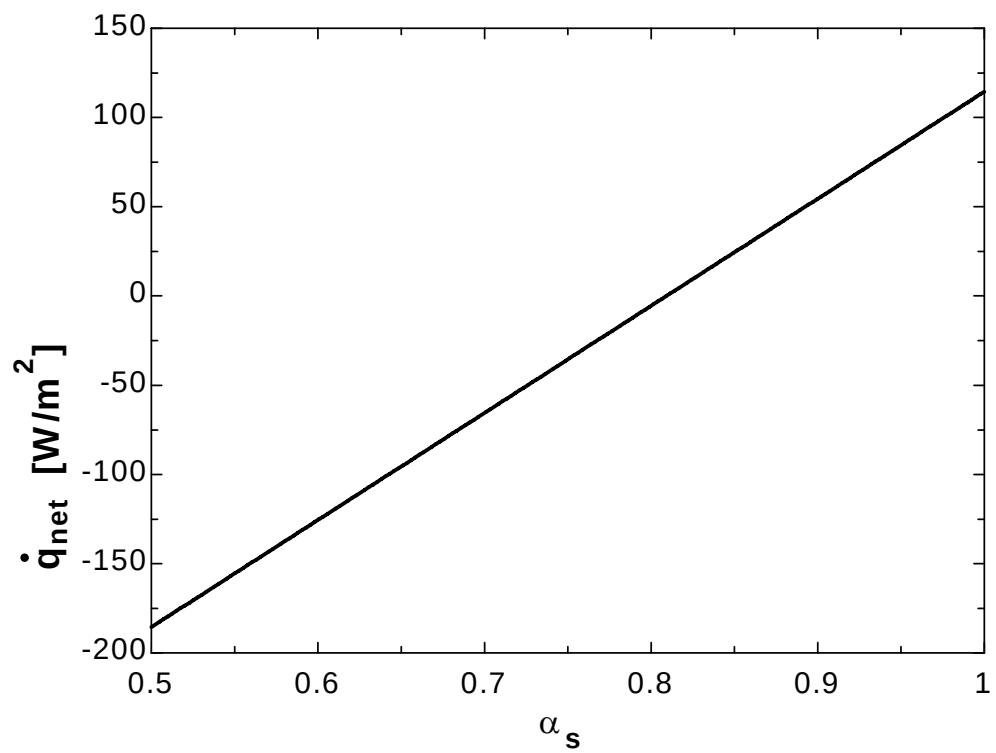
"ANALYSIS"

$q_{\text{dot_net}}=q_{\text{dot_gain}}-q_{\text{dot_loss}}$ "energy balance"

$q_{\text{dot_gain}}=\alpha_s G_{\text{solar}}$

$q_{\text{dot_loss}}=\epsilon\sigma(T_s^4-T_{\text{sky}}^4)+h(T_s-T_{\text{air}})$

α_s	$q_{\text{net}} \text{ [W/m}^2\text{]}$
0.5	-185.5
0.525	-170.5
0.55	-155.5
0.575	-140.5
0.6	-125.5
0.625	-110.5
0.65	-95.52
0.675	-80.52
0.7	-65.52
0.725	-50.52
0.75	-35.52
0.775	-20.52
0.8	-5.525
0.825	9.475
0.85	24.48
0.875	39.48
0.9	54.48
0.925	69.48
0.95	84.48
0.975	99.48
1	114.5



11-63 The absorber surface of a solar collector is exposed to solar and sky radiation. The equilibrium temperature of the absorber surface is to be determined if the backside of the plate is insulated.

Properties The solar absorptivity and emissivity of the surface are given to $\alpha_s = 0.87$ and $\varepsilon = 0.09$.

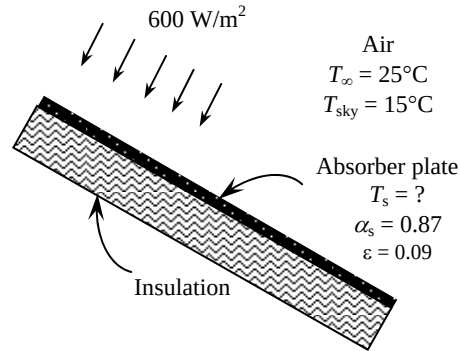
Analysis The backside of the absorbing plate is insulated (instead of being attached to water tubes), and thus

$$\dot{q}_{net} = 0$$

$$\alpha_s G_{solar} = \varepsilon \sigma (T_s^4 - T_{sky}^4) + h(T_s - T_{air})$$

$$(0.87)(600 \text{ W/m}^2) = (0.09)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(T_s)^4 - (288 \text{ K})^4] + (10 \text{ W/m}^2 \cdot \text{K})(T_s - 298 \text{ K})$$

$$T_s = \mathbf{346 \text{ K}}$$



Special Topic: Solar Heat Gain Through Windows

11-64C (a) The spectral distribution of solar radiation beyond the earth's atmosphere resembles the energy emitted by a black body at 5982°C, with about 39 percent in the visible region (0.4 to 0.7 μm), and the 52 percent in the near infrared region (0.7 to 3.5 μm). (b) At a solar altitude of 41.8°, the total energy of direct solar radiation incident at sea level on a clear day consists of about 3 percent ultraviolet, 38 percent visible, and 59 percent infrared radiation.

11-65C A window that transmits visible part of the spectrum while absorbing the infrared portion is ideally suited for minimizing the air-conditioning load since such windows provide maximum daylighting and minimum solar heat gain. The ordinary window glass approximates this behavior remarkably well.

11-66C The **solar heat gain coefficient (SHGC)** is defined as the fraction of incident solar radiation that enters through the glazing. The solar heat gain of a glazing relative to the solar heat gain of a reference glazing, typically that of a standard 3 mm (1/8 in) thick double-strength clear window glass sheet whose SHGC is 0.87, is called the **shading coefficient**. They are related to each other by

$$\text{SC} = \frac{\text{Solar heat gain of product}}{\text{Solar heat gain of reference glazing}} = \frac{\text{SHGC}}{\text{SHGC}_{\text{ref}}} = \frac{\text{SHGC}}{0.87} = 1.15 \times \text{SHGC}$$

For single pane clear glass window, SHGC = 0.87 and SC = 1.0.

11-67C The SC (shading coefficient) of a device represents the solar heat gain relative to the solar heat gain of a reference glazing, typically that of a standard 3 mm (1/8 in) thick double-strength clear window glass sheet whose SHGC is 0.87. The shading coefficient of a 3-mm thick *clear glass* is SC = 1.0 whereas SC = 0.88 for 3-mm thick *heat absorbing glass*.

11-68C A device that blocks solar radiation and thus reduces the solar heat gain is called a shading device. External shading devices are more effective in reducing the solar heat gain since they intercept sun's rays before they reach the glazing. The solar heat gain through a window can be reduced by as much as 80 percent by exterior shading. *Light colored* shading devices maximize the back reflection and thus minimize the solar gain. *Dark colored* shades, on the other hand, minimize the back reflection and thus maximize the solar heat gain.

11-69C A low-e coating on the inner surface of a window glass reduces both the (a) heat loss in winter and (b) heat gain in summer. This is because the radiation heat transfer to or from the window is proportional to the emissivity of the inner surface of the window. In winter, the window is colder and thus radiation heat loss from the room to the window is low. In summer, the window is hotter and the radiation transfer from the window to the room is low.

11-70C Glasses coated with reflective films on the outer surface of a window glass reduces solar heat both in summer and in winter.

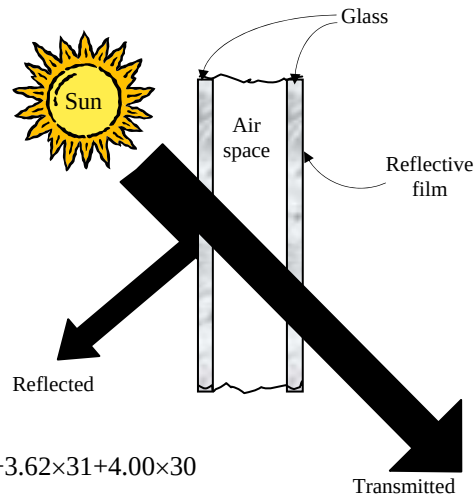
11-71 The net annual cost savings due to installing reflective coating on the West windows of a building and the simple payback period are to be determined.

Assumptions 1 The calculations given below are for an average year. 2 The unit costs of electricity and natural gas remain constant.

Analysis Using the daily averages for each month and noting the number of days of each month, the total solar heat flux incident on the glazing during summer and winter months are determined to be

$$Q_{\text{solar, summer}} = 4.24 \times 30 + 4.16 \times 31 + 3.93 \times 31 + 3.48 \times 30 \\ = 482 \text{ kWh/year}$$

$$Q_{\text{solar, winter}} = 2.94 \times 31 + 2.33 \times 30 + 2.07 \times 31 + 2.35 \times 31 + 3.03 \times 28 + 3.62 \times 31 + 4.00 \times 30 \\ = 615 \text{ kWh/year}$$



Then the decrease in the annual cooling load and the increase in the annual heating load due to reflective film become

$$\text{Cooling load decrease} = Q_{\text{solar, summer}} A_{\text{glazing}} (\text{SHGC}_{\text{without film}} - \text{SHGC}_{\text{with film}}) \\ = (482 \text{ kWh/year})(60 \text{ m}^2)(0.766 - 0.261) \\ = 14,605 \text{ kWh/year}$$

$$\text{Heating load increase} = Q_{\text{solar, winter}} A_{\text{glazing}} (\text{SHGC}_{\text{without film}} - \text{SHGC}_{\text{with film}}) \\ = (615 \text{ kWh/year})(60 \text{ m}^2)(0.766 - 0.261) \\ = 18,635 \text{ kWh/year} = 635.8 \text{ therms/year}$$

since 1 therm = 29.31 kWh. The corresponding decrease in cooling costs and increase in heating costs are

$$\text{Decrease in cooling costs} = (\text{Cooling load decrease})(\text{Unit cost of electricity})/\text{COP} \\ = (14,605 \text{ kWh/year})(\$0.09/\text{kWh})/3.2 = \$411/\text{year}$$

$$\text{Increase in heating costs} = (\text{Heating load increase})(\text{Unit cost of fuel})/\text{Efficiency} \\ = (635.8 \text{ therms/year})(\$0.45/\text{therm})/0.80 = \$358/\text{year}$$

Then the net annual cost savings due to reflective films become

$$\text{Cost Savings} = \text{Decrease in cooling costs} - \text{Increase in heating costs} = \$411 - 358 = \$53/\text{year}$$

The implementation cost of installing films is

$$\text{Implementation Cost} = (\$20/\text{m}^2)(60 \text{ m}^2) = \$1200$$

This gives a simple payback period of

$$\text{Simple payback period} = \frac{\text{Implementation cost}}{\text{Annual cost savings}} = \frac{\$1200}{\$53/\text{year}} = \mathbf{23 \text{ years}}$$

Discussion The reflective films will pay for themselves in this case in about 23 years, which is unacceptable to most manufacturers since they are not usually interested in any energy conservation measure which does not pay for itself within 3 years.

11-72 A house located at 40° N latitude has ordinary double pane windows. The total solar heat gain of the house at 9:00, 12:00, and 15:00 solar time in July and the total amount of solar heat gain per day for an average day in January are to be determined.

Assumptions The calculations are performed for an average day in a given month.

Properties The shading coefficient of a double pane window with 6-mm thick glasses is $SC = 0.82$ (Table 11-5). The incident radiation at different windows at different times are given as (Table 11-4)

Month	Time	Solar radiation incident on the surface, W/m^2			
		North	East	South	West
July	9:00	117	701	190	114
July	12:00	138	149	395	149
July	15:00	117	114	190	701
January	Daily total	446	1863	5897	1863

Analysis The solar heat gain coefficient (SHGC) of the windows is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 0.82 = 0.7134$$

The rate of solar heat gain is determined from

$$\dot{Q}_{\text{solar gain}} = SHGC \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}} = 0.7134 \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}}$$

Then the rates of heat gain at the 4 walls at 3 different times in July become

North wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.7134 \times (4 \text{ m}^2) \times (117 \text{ W/m}^2) = \mathbf{334 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.7134 \times (4 \text{ m}^2) \times (138 \text{ W/m}^2) = \mathbf{394 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.7134 \times (4 \text{ m}^2) \times (117 \text{ W/m}^2) = \mathbf{334 \text{ W}}\end{aligned}$$

East wall:

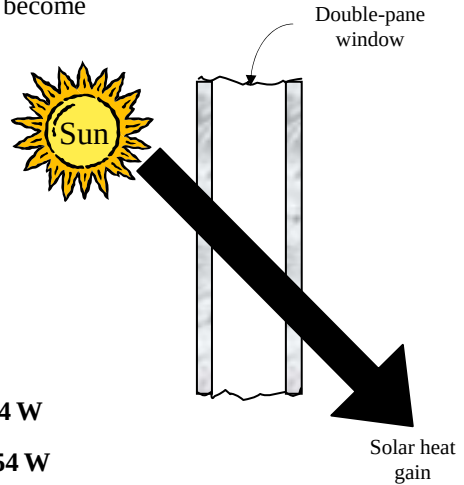
$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.7134 \times (6 \text{ m}^2) \times (701 \text{ W/m}^2) = \mathbf{3001 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.7134 \times (6 \text{ m}^2) \times (149 \text{ W/m}^2) = \mathbf{638 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.7134 \times (6 \text{ m}^2) \times (114 \text{ W/m}^2) = \mathbf{488 \text{ W}}\end{aligned}$$

South wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.7134 \times (8 \text{ m}^2) \times (190 \text{ W/m}^2) = \mathbf{1084 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.7134 \times (8 \text{ m}^2) \times (395 \text{ W/m}^2) = \mathbf{2254 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.7134 \times (8 \text{ m}^2) \times (190 \text{ W/m}^2) = \mathbf{1084 \text{ W}}\end{aligned}$$

West wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.7134 \times (6 \text{ m}^2) \times (114 \text{ W/m}^2) = \mathbf{488 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.7134 \times (6 \text{ m}^2) \times (149 \text{ W/m}^2) = \mathbf{638 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.7134 \times (6 \text{ m}^2) \times (701 \text{ W/m}^2) = \mathbf{3001 \text{ W}}\end{aligned}$$



Similarly, the solar heat gain of the house through all of the windows in January is determined to be

January:

$$\begin{aligned}\dot{Q}_{\text{solar gain, North}} &= 0.7134 \times (4 \text{ m}^2) \times (446 \text{ Wh/m}^2 \cdot \text{day}) = 1273 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, East}} &= 0.7134 \times (6 \text{ m}^2) \times (1863 \text{ Wh/m}^2 \cdot \text{day}) = 7974 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, South}} &= 0.7134 \times (8 \text{ m}^2) \times (5897 \text{ Wh/m}^2 \cdot \text{day}) = 33,655 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, West}} &= 0.7134 \times (6 \text{ m}^2) \times (1863 \text{ Wh/m}^2 \cdot \text{day}) = 7974 \text{ Wh/day}\end{aligned}$$

Therefore, for an average day in January,

$$\dot{Q}_{\text{solar gain per day}} = 446 + 1863 + 5897 + 1863 = 58,876 \text{ Wh/day} \cong \mathbf{58.9 \text{ kWh/day}}$$

11-73 A house located at 40° N latitude has gray-tinted double pane windows. The total solar heat gain of the house at 9:00, 12:00, and 15:00 solar time in July and the total amount of solar heat gain per day for an average day in January are to be determined.

Assumptions The calculations are performed for an average day in a given month.

Properties The shading coefficient of a gray-tinted double pane window with 6-mm thick glasses is $SC = 0.58$ (Table 11-5). The incident radiation at different windows at different times are given as (Table 11-4)

Month	Time	Solar radiation incident on the surface, W/m^2			
		North	East	South	West
July	9:00	117	701	190	114
July	12:00	138	149	395	149
July	15:00	117	114	190	701
January	Daily total	446	1863	5897	1863

Analysis The solar heat gain coefficient (SHGC) of the windows is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 0.58 = 0.5046$$

The rate of solar heat gain is determined from

$$\dot{Q}_{\text{solar gain}} = SHGC \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}} = 0.5046 \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}}$$

Then the rates of heat gain at the 4 walls at 3 different times in July become

North wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.5046 \times (4 \text{ m}^2) \times (117 \text{ W/m}^2) = \mathbf{236 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.5046 \times (4 \text{ m}^2) \times (138 \text{ W/m}^2) = \mathbf{279 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.5046 \times (4 \text{ m}^2) \times (117 \text{ W/m}^2) = \mathbf{236 \text{ W}}\end{aligned}$$

East wall:

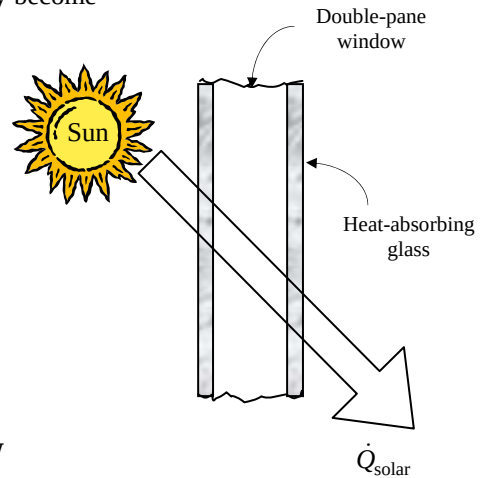
$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.5046 \times (6 \text{ m}^2) \times (701 \text{ W/m}^2) = \mathbf{2122 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.5046 \times (6 \text{ m}^2) \times (149 \text{ W/m}^2) = \mathbf{461 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.5046 \times (6 \text{ m}^2) \times (114 \text{ W/m}^2) = \mathbf{345 \text{ W}}\end{aligned}$$

South wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.5046 \times (8 \text{ m}^2) \times (190 \text{ W/m}^2) = \mathbf{7674 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.5046 \times (8 \text{ m}^2) \times (395 \text{ W/m}^2) = \mathbf{1595 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.5046 \times (8 \text{ m}^2) \times (190 \text{ W/m}^2) = \mathbf{767 \text{ W}}\end{aligned}$$

West wall:

$$\begin{aligned}\dot{Q}_{\text{solar gain, 9:00}} &= 0.5046 \times (6 \text{ m}^2) \times (114 \text{ W/m}^2) = \mathbf{345 \text{ W}} \\ \dot{Q}_{\text{solar gain, 12:00}} &= 0.5046 \times (6 \text{ m}^2) \times (149 \text{ W/m}^2) = \mathbf{451 \text{ W}} \\ \dot{Q}_{\text{solar gain, 15:00}} &= 0.5046 \times (6 \text{ m}^2) \times (701 \text{ W/m}^2) = \mathbf{2122 \text{ W}}\end{aligned}$$



Similarly, the solar heat gain of the house through all of the windows in January is determined to be

January:

$$\begin{aligned}\dot{Q}_{\text{solar gain, North}} &= 0.5046 \times (4 \text{ m}^2) \times (446 \text{ Wh/m}^2 \cdot \text{day}) = 900 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, East}} &= 0.5046 \times (6 \text{ m}^2) \times (1863 \text{ Wh/m}^2 \cdot \text{day}) = 5640 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, South}} &= 0.5046 \times (8 \text{ m}^2) \times (5897 \text{ Wh/m}^2 \cdot \text{day}) = 23,805 \text{ Wh/day} \\ \dot{Q}_{\text{solar gain, West}} &= 0.5046 \times (6 \text{ m}^2) \times (1863 \text{ Wh/m}^2 \cdot \text{day}) = 5640 \text{ Wh/day}\end{aligned}$$

Therefore, for an average day in January,

$$\dot{Q}_{\text{solar gain per day}} = 900 + 5640 + 23,805 + 5640 = 35,985 \text{ Wh/day} = \mathbf{35.895 \text{ kWh/day}}$$

11-74 A building at 40° N latitude has double pane heat absorbing type windows that are equipped with light colored venetian blinds. The total solar heat gains of the building through the south windows at solar noon in April for the cases of with and without the blinds are to be determined.

Assumptions The calculations are performed for an “average” day in April, and may vary from location to location.

Properties The shading coefficient of a double pane heat absorbing type windows is $SC = 0.58$ (Table 11-5). It is given to be $SC = 0.30$ in the case of blinds. The solar radiation incident at a South-facing surface at 12:00 noon in April is 559 W/m^2 (Table 11-4).

Analysis The solar heat gain coefficient (SHGC) of the windows without the blinds is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 0.58 = 0.5046$$

Then the rate of solar heat gain through the window becomes

$$\begin{aligned}\dot{Q}_{\text{solar gain, no blinds}} &= SHGC \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}} \\ &= 0.5046(200 \text{ m}^2)(559 \text{ W/m}^2) \\ &= \mathbf{56,414 \text{ W}}\end{aligned}$$

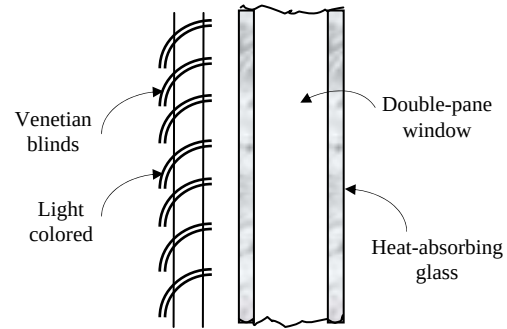
In the case of windows equipped with venetian blinds, the SHGC and the rate of solar heat gain become

$$SHGC = 0.87 \times SC = 0.87 \times 0.30 = 0.261$$

Then the rate of solar heat gain through the window becomes

$$\begin{aligned}\dot{Q}_{\text{solar gain, no blinds}} &= SHGC \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}} \\ &= 0.261(200 \text{ m}^2)(559 \text{ W/m}^2) \\ &= \mathbf{29,180 \text{ W}}\end{aligned}$$

Discussion Note that light colored venetian blinds significantly reduce the solar heat, and thus air-conditioning load in summers.



11-75 A house has double door type windows that are double pane with 6.4 mm of air space and aluminum frames and spacers. It is to be determined if the house is losing more or less heat than it is gaining from the sun through an east window in a typical day in January.

Assumptions 1 The calculations are performed for an “average” day in January. 2 Solar data at 40° latitude can also be used for a location at 39° latitude.

Properties The shading coefficient of a double pane window with 3-mm thick clear glass is $SC = 0.88$ (Table 11-5). The overall heat transfer coefficient for double door type windows that are double pane with 6.4 mm of air space and aluminum frames and spacers is $4.55 \text{ W/m}^2 \cdot ^\circ\text{C}$. (Table 9-6). The total solar radiation incident at an East-facing surface in January during a typical day is 1863 Wh/m^2 (Table 11-4).

Analysis The solar heat gain coefficient (SHGC) of the windows is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 0.88 = 0.7656$$

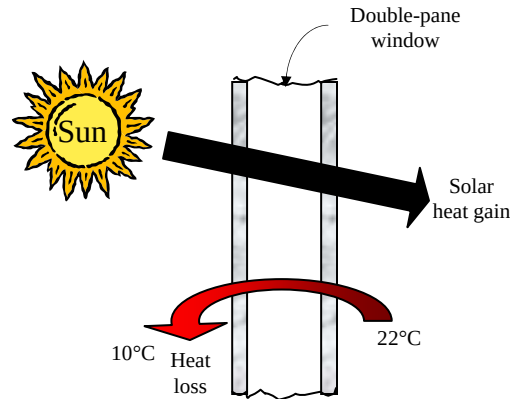
Then the solar heat gain through the window per unit area becomes

$$\begin{aligned} Q_{\text{solar gain}} &= SHGC \times A_{\text{glazing}} \times q_{\text{solar, daily total}} \\ &= 0.7656(1 \text{ m}^2)(1863 \text{ Wh/m}^2) \\ &= \mathbf{1426 \text{ Wh} = 1.426 \text{ kWh}} \end{aligned}$$

The heat loss through a unit area of the window during a 24-h period is

$$\begin{aligned} Q_{\text{loss, window}} &= \dot{Q}_{\text{loss, window}} \Delta t = U_{\text{window}} A_{\text{window}} (T_i - T_{0, \text{ave}})(1 \text{ day}) \\ &= (4.55 \text{ W/m}^2 \cdot ^\circ\text{C})(1 \text{ m}^2)(22 - 10)^\circ\text{C}(24 \text{ h}) \\ &= \mathbf{1310 \text{ Wh} = 1.31 \text{ kWh}} \end{aligned}$$

Therefore, the house is **gaining** more heat than it is losing through the East windows during a typical day in January.



11-76 A house has double door type windows that are double pane with 6.4 mm of air space and aluminum frames and spacers. It is to be determined if the house is losing more or less heat than it is gaining from the sun through a South window in a typical day in January.

Assumptions 1 The calculations are performed for an “average” day in January. 2 Solar data at 40° latitude can also be used for a location at 39° latitude.

Properties The shading coefficient of a double pane window with 3-mm thick clear glass is $SC = 0.88$ (Table 11-5). The overall heat transfer coefficient for double door type windows that are double pane with 6.4 mm of air space and aluminum frames and spacers is $4.55 \text{ W/m}^2 \cdot ^\circ\text{C}$ (Table 9-6). The total solar radiation incident at a South-facing surface in January during a typical day is 5897 Wh/m^2 (Table 11-5).

Analysis The solar heat gain coefficient (SHGC) of the windows is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 0.88 = 0.7656$$

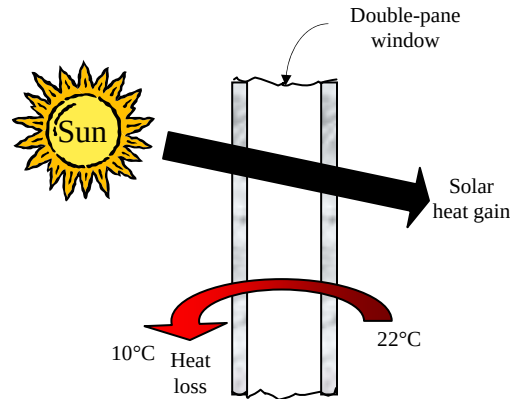
Then the solar heat gain through the window per unit area becomes

$$\begin{aligned} Q_{\text{solar gain}} &= SHGC \times A_{\text{glazing}} \times q_{\text{solar, daily total}} \\ &= 0.7656(1 \text{ m}^2)(5897 \text{ Wh/m}^2) \\ &= \mathbf{4515 \text{ Wh} = 4.515 \text{ kWh}} \end{aligned}$$

The heat loss through a unit area of the window during a 24-h period is

$$\begin{aligned} Q_{\text{loss, window}} &= \dot{Q}_{\text{loss, window}} \Delta t = U_{\text{window}} A_{\text{window}} (T_i - T_{o, \text{ave}})(1 \text{ day}) \\ &= (4.55 \text{ W/m}^2 \cdot ^\circ\text{C})(1 \text{ m}^2)(22 - 10)^\circ\text{C}(24 \text{ h}) \\ &= \mathbf{1310 \text{ Wh} = 1.31 \text{ kWh}} \end{aligned}$$

Therefore, the house is **gaining** much more heat than it is losing through the South windows during a typical day in January.



11-77E A house has 1/8-in thick single pane windows with aluminum frames on a West wall. The rate of net heat gain (or loss) through the window at 3 PM during a typical day in January is to be determined.

Assumptions 1 The calculations are performed for an “average” day in January. 2 The frame area relative to glazing area is small so that the glazing area can be taken to be the same as the window area.

Properties The shading coefficient of a 1/8-in thick single pane window is $SC = 1.0$ (Table 11-5). The overall heat transfer coefficient for 1/8-in thick single pane windows with aluminum frames is $6.63 \text{ W/m}^2 \cdot ^\circ\text{C} = 1.17 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$ (Table 9-6). The total solar radiation incident at a West-facing surface at 3 PM in January during a typical day is $557 \text{ W/m}^2 = 177 \text{ Btu/h} \cdot \text{ft}^2$ (Table 11-4).

Analysis The solar heat gain coefficient (SHGC) of the windows is determined from Eq.11-57 to be

$$SHGC = 0.87 \times SC = 0.87 \times 1.0 = 0.87$$

The window area is: $A_{\text{window}} = (9 \text{ ft})(15 \text{ ft}) = 135 \text{ ft}^2$

Then the rate of solar heat gain through the window at 3 PM becomes

$$\begin{aligned}\dot{Q}_{\text{solar gain, 3 PM}} &= SHGC \times A_{\text{glazing}} \times \dot{q}_{\text{solar, 3 PM}} \\ &= 0.87(135 \text{ ft}^2)(177 \text{ Btu/h} \cdot \text{ft}^2) \\ &= 20,789 \text{ Btu/h}\end{aligned}$$

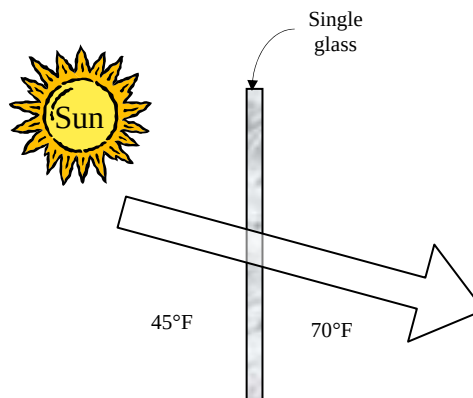
The rate of heat loss through the window at 3 PM is

$$\begin{aligned}\dot{Q}_{\text{loss, window}} &= U_{\text{window}} A_{\text{window}} (T_i - T_o) \\ &= (1.17 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(135 \text{ ft}^2)(70 - 45)^\circ\text{F} \\ &= 3949 \text{ Btu/h}\end{aligned}$$

The house will be gaining heat at 3 PM since the solar heat gain is larger than the heat loss. The rate of net heat gain through the window is

$$\dot{Q}_{\text{net}} = \dot{Q}_{\text{solar gain, 3 PM}} - \dot{Q}_{\text{loss, window}} = 20,789 - 394 = \mathbf{16,840 \text{ Btu/h}}$$

Discussion The actual heat gain will be less because of the area occupied by the window frame.



11-78 A building located near 40° N latitude has equal window areas on all four sides. The side of the building with the highest solar heat gain in summer is to be determined.

Assumptions The shading coefficients of windows on all sides of the building are identical.

Analysis The reflective films should be installed on the side that receives the most incident solar radiation in summer since the window areas and the shading coefficients on all four sides are identical. The incident solar radiation at different windows in July are given to be (Table 11-5)

Month	Time	The daily total solar radiation incident on the surface, Wh/m^2			
		North	East	South	West
July	Daily total	1621	4313	2552	4313

Therefore, the reflective film should be installed on the **East** or **West** windows (instead of the South windows) in order to minimize the solar heat gain and thus the cooling load of the building.

Review Problems

11-79 The variation of emissivity of an opaque surface at a specified temperature with wavelength is given. The average emissivity of the surface and its emissive power are to be determined.

Analysis The average emissivity of the surface can be determined from

$$\varepsilon(T) = \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (f_{\lambda_2} - f_{\lambda_1}) + \varepsilon_3 (1 - f_{\lambda_2})$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$. These functions are determined from Table 11-1 to be

$$\lambda_1 T = (2 \mu\text{m})(1200 \text{ K}) = 2400 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.140256$$

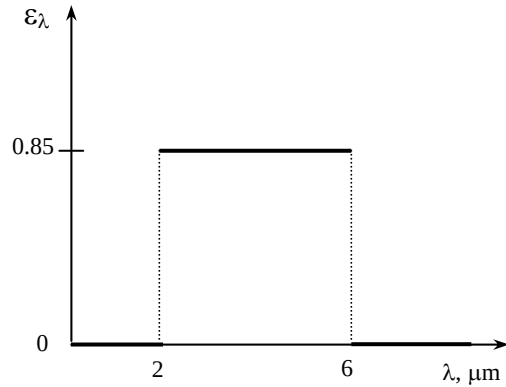
$$\lambda_2 T = (6 \mu\text{m})(1200 \text{ K}) = 7200 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.819217$$

and

$$\varepsilon = (0.0)(0.140256) + (0.85)(0.819217 - 0.140256) + (0.0)(1 - 0.819217) = \mathbf{0.577}$$

Then the emissive flux of the surface becomes

$$E = \varepsilon \sigma T^4 = (0.577)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1200 \text{ K})^4 = \mathbf{67,853 \text{ W/m}^2}$$



11-80 The variation of transmissivity of glass with wavelength is given. The transmissivity of the glass for solar radiation and for light are to be determined.

Analysis For solar radiation, $T = 5800 \text{ K}$. The average transmissivity of the surface can be determined from

$$\tau(T) = \tau_1 f_{\lambda_1} + \tau_2 (f_{\lambda_2} - f_{\lambda_1}) + \tau_3 (1 - f_{\lambda_2})$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$. These functions are determined from Table 11-1 to be

$$\lambda_1 T = (0.35 \mu\text{m})(5800 \text{ K}) = 2030 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.071852$$

$$\lambda_2 T = (2.5 \mu\text{m})(5800 \text{ K}) = 14,500 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.966440$$

and

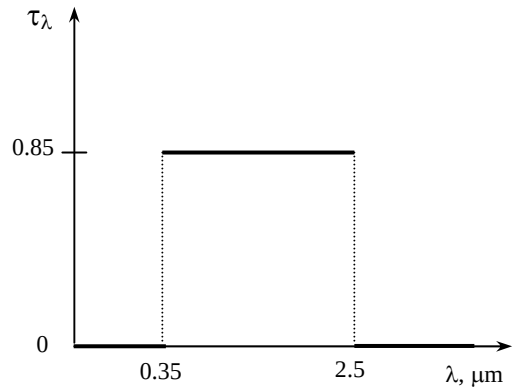
$$\tau = (0.0)(0.071852) + (0.85)(0.966440 - 0.071852) + (0.0)(1 - 0.966440) = \mathbf{0.760}$$

For light, we take $T = 300 \text{ K}$. Repeating the calculations at this temperature we obtain

$$\lambda_1 T = (0.35 \mu\text{m})(300 \text{ K}) = 105 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.00$$

$$\lambda_2 T = (2.5 \mu\text{m})(300 \text{ K}) = 750 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.000012$$

$$\tau = (0.0)(0.00) + (0.85)(0.000012 - 0.00) + (0.0)(1 - 0.000012) = \mathbf{0.00001}$$



11-81 A hole is drilled in a spherical cavity. The maximum rate of radiation energy streaming through the hole is to be determined.

Analysis The maximum rate of radiation energy streaming through the hole is the blackbody radiation, and it can be determined from

$$E = A\sigma T^4 = \pi(0.0025 \text{ m})^2 (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(600 \text{ K})^4 = \mathbf{0.144 \text{ W}}$$

The result would not change for a different diameter of the cavity.

11-82 The variation of absorptivity of a surface with wavelength is given. The average absorptivity of the surface is to be determined for two source temperatures.

Analysis (a) $T = 1000 \text{ K}$. The average absorptivity of the surface can be determined from

$$\begin{aligned} \alpha(T) &= \alpha_1 f_{0-\lambda_1} + \alpha_2 f_{\lambda_1-\lambda_2} + \alpha_3 f_{\lambda_2-\infty} \\ &= \alpha_1 f_{\lambda_1} + \alpha_2 (f_{\lambda_2} - f_{\lambda_1}) + \alpha_3 (1 - f_{\lambda_2}) \end{aligned}$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$, determined from

$$\lambda_1 T = (0.3 \mu\text{m})(1000 \text{ K}) = 300 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.0$$

$$\lambda_2 T = (1.2 \mu\text{m})(1000 \text{ K}) = 1200 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.002134$$

$$f_{0-\lambda_1} = f_{\lambda_1} - f_0 = f_{\lambda_1} \text{ since } f_0 = 0 \text{ and } f_{\lambda_2-\infty} = f_{\infty} - f_{\lambda_2} \text{ since } f_{\infty} = 1.$$

and,

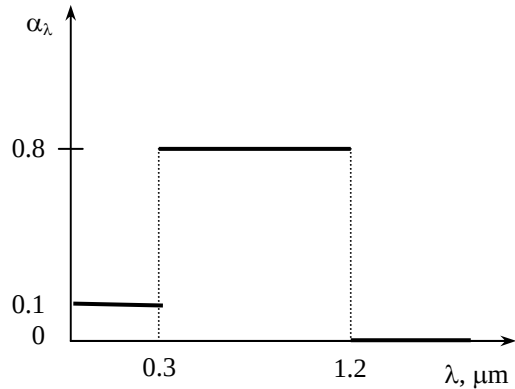
$$\alpha = (0.1)0.0 + (0.8)(0.002134 - 0.0) + (0.0)(1 - 0.002134) = \mathbf{0.0017}$$

(a) $T = 3000 \text{ K}$.

$$\lambda_1 T = (0.3 \mu\text{m})(3000 \text{ K}) = 900 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.000169$$

$$\lambda_2 T = (1.2 \mu\text{m})(3000 \text{ K}) = 3600 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.403607$$

$$\alpha = (0.1)0.000169 + (0.8)(0.403607 - 0.000169) + (0.0)(1 - 0.403607) = \mathbf{0.323}$$



11-83 The variation of absorptivity of a surface with wavelength is given. The surface receives solar radiation at a specified rate. The solar absorptivity of the surface and the rate of absorption of solar radiation are to be determined.

Analysis For solar radiation, $T = 5800$ K. The solar absorptivity of the surface is

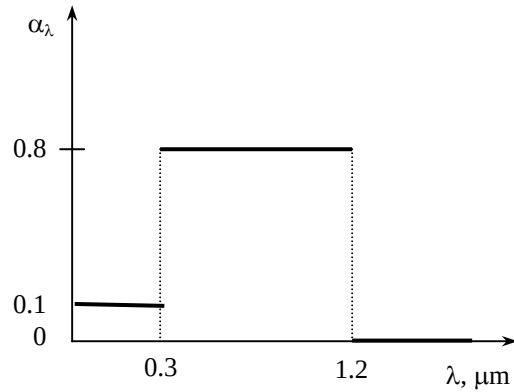
$$\lambda_1 T = (0.3 \mu\text{m})(5800 \text{ K}) = 1740 \mu\text{mK} \rightarrow f_{\lambda_1} = 0.033454$$

$$\lambda_2 T = (1.2 \mu\text{m})(5800 \text{ K}) = 6960 \mu\text{mK} \rightarrow f_{\lambda_2} = 0.805713$$

$$\begin{aligned} \alpha &= (0.1)(0.033454) + (0.8)(0.805713 - 0.033454) \\ &\quad + (0.0)(1 - 0.805713) \\ &= \mathbf{0.621} \end{aligned}$$

The rate of absorption of solar radiation is determined from

$$E_{\text{absorbed}} = \alpha I = 0.621(820 \text{ W/m}^2) = \mathbf{509 \text{ W/m}^2}$$



11-84 The spectral transmissivity of a glass cover used in a solar collector is given. Solar radiation is incident on the collector. The solar flux incident on the absorber plate, the transmissivity of the glass cover for radiation emitted by the absorber plate, and the rate of heat transfer to the cooling water are to be determined.

Analysis (a) For solar radiation, $T = 5800$ K. The average transmissivity of the surface can be determined from

$$\tau(T) = \tau_1 f_{\lambda_1} + \tau_2 (f_{\lambda_2} - f_{\lambda_1}) + \tau_3 (1 - f_{\lambda_2})$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$. These functions are determined from Table 11-1 to be

$$\lambda_1 T = (0.3 \mu\text{m})(5800 \text{ K}) = 1740 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.033454$$

$$\lambda_2 T = (3 \mu\text{m})(5800 \text{ K}) = 17,400 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.978746$$

and

$$\tau = (0.0)(0.033454) + (0.9)(0.978746 - 0.033454) + (0.0)(1 - 0.978746) = 0.851$$

Since the absorber plate is black, all of the radiation transmitted through the glass cover will be absorbed by the absorber plate and therefore, the solar flux incident on the absorber plate is same as the radiation absorbed by the absorber plate:

$$E_{\text{abs. plate}} = \tau I = 0.851(950 \text{ W/m}^2) = \mathbf{808.5 \text{ W/m}^2}$$

(b) For radiation emitted by the absorber plate, we take $T = 300$ K, and calculate the transmissivity as follows:

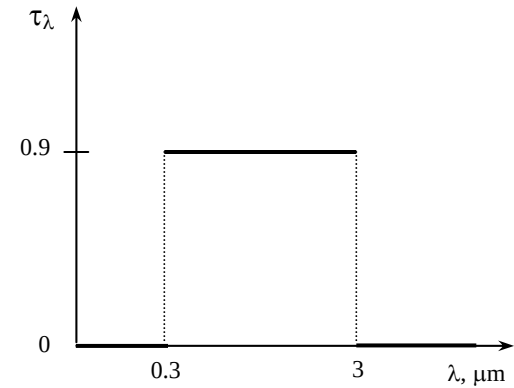
$$\lambda_1 T = (0.3 \mu\text{m})(300 \text{ K}) = 90 \mu\text{mK} \longrightarrow f_{\lambda_1} = 0.0$$

$$\lambda_2 T = (3 \mu\text{m})(300 \text{ K}) = 900 \mu\text{mK} \longrightarrow f_{\lambda_2} = 0.000169$$

$$\tau = (0.0)(0.0) + (0.9)(0.000169 - 0.0) + (0.0)(1 - 0.000169) = \mathbf{0.00015}$$

(c) The rate of heat transfer to the cooling water is the difference between the radiation absorbed by the absorber plate and the radiation emitted by the absorber plate, and it is determined from

$$\dot{Q}_{\text{water}} = (\tau_{\text{solar}} - \tau_{\text{room}}) I = (0.851 - 0.00015)(950 \text{ W/m}^2) = \mathbf{808.3 \text{ W/m}^2}$$



11-85 A small surface emits radiation. The rate of radiation energy emitted through a band is to be determined.

Assumptions Surface A emits diffusely as a blackbody.

Analysis The rate of radiation emission from a surface per unit surface area in the direction (θ, ϕ) is given as

$$dE = \frac{d\dot{Q}_e}{dA} = I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi$$

The total rate of radiation emission through the band between 40° and 50° can be expressed as

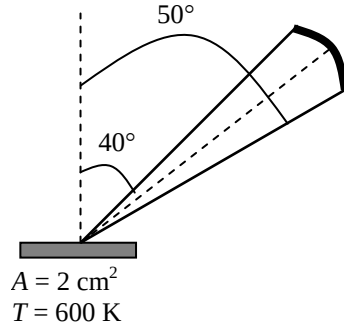
$$E = \int_{\phi=0}^{2\pi} \int_{\theta=40}^{50} I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi = I_b (0.1736\pi) = \frac{\sigma T^4}{\pi} (0.1736\pi) = 0.1736\sigma T^4$$

since the blackbody radiation intensity is constant ($I_b = \text{constant}$), and

$$\int_{\phi=0}^{2\pi} \int_{\theta=40}^{50} \cos \theta \sin \theta d\theta d\phi = 2\pi \int_{\theta=40}^{50} \cos \theta \sin \theta d\theta = \pi(\sin^2 50 - \sin^2 40) = 0.1736\pi$$

Approximating a small area as a differential area, the rate of radiation energy emitted from an area of 1 cm^2 in the specified band becomes

$$\dot{Q}_e = E dA = 0.1736\sigma T^4 dA = 0.1736 \times (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) (600 \text{ K})^4 (1 \times 10^{-4} \text{ m}^2) = \mathbf{0.128 \text{ W}}$$



11-86 11-87 Design and Essay Problems
